Chapter 1

Introduction

This chapter introduces the history of the HealLM project, the motivation to create an intelligent mental health assistance system, and the drawbacks of the current approaches. It also outlines the aims of the project and summarizes the proposed methodology and the expected outcomes of the project.

1.1 Introduction

Mental health is a fundamental part of health and wellbeing. It affects people's ability to manage everyday events, form relationships, learn, work and engage in society. Mental health disorders are a significant public-health problem worldwide. In 2021, the World Health Organization (WHO) estimates that about 1.1 billion people, or nearly one in seven of the global population, suffered a mental disorder, with anxiety and depressive disorders among the most common disorders [1]. Most individuals with mental disorders are not receiving effective treatment, although there are effective options available for prevention and treatment.

One of the most common types of mental disorders is anxiety disorder. As of 2021, WHO estimated that there were 359 million people with an anxiety disorder in the world, making anxiety disorders the most common mental disorder in the world [2]. Symptoms of anxiety can include feeling anxious or worried to the point where it is out of control, concentrating less than normal, getting irritated or angry easily, feeling restless, sleeping problems, and physical issues like palpitations and sweating. There are effective treatments available, but about 1 in 4 people with an anxiety disorder receive treatment.

Depression is another major mental health condition affecting individuals across different populations. WHO estimates that approximately 332 million people worldwide experience depression [3]. Depression can involve persistent sadness, loss of interest or pleasure, sleep and appetite disturbances, fatigue, poor concentration, feelings of worthlessness, and hopelessness. The condition can substantially affect an individual's personal, social, academic, and occupational functioning.

The situation is especially critical in Bangladesh. WHO's information on mental health in Bangladesh indicates that over 6 million people in Bangladesh suffer from depressive disorders and nearly 7 million suffer from anxiety disorders [4]. The mental health system also has significant barriers to accessing professional services in the country.

The overall mental-health scenario in Bangladesh remains one of significant treatment gaps, especially for anxiety and depressive disorders. In a recent nationally representative study of 7270 Bangladeshi adults, 3.5% were diagnosed with Generalized Anxiety Disorder (GAD), and among them only 3.9% people visited mental health services, highlighting the gap between mental health needs and services [5]. The study emphasizes the importance of mental-health awareness programs in the community, the need for the mental-health workforce to be strengthened in Bangladesh and making mental-health services part of primary care.

A significant feature of mental health issues is that various issues can happen at the same time, and the indicators of multiple mental-health conditions can overlap. Symptoms may include sleep disturbances, concentration problems, fatigue, irritability, or changes in mood and are heightened by anxiety and depression. Therefore, it is crucial to know how symptoms relate to multiple mental-health conditions so that appropriate information regarding mental-health conditions is provided. WHO also reports that anxiety disorders are associated with an increased risk of depression and that people can have more than one anxiety disorder at the same time [6].

The advent of new technologies like Artificial Intelligence (AI) and Large Language Models (LLMs) has presented new possibilities for healthcare and mental-health conversational systems. LLMs can handle natural-language descriptions and produce human-like responses, which could be beneficial for counselling, emotional support, clinical aid, and disseminating mental-health information. In the 2025 scoping review of LLM applications in mental-health care, there were various applications found, including clinical, counselling, therapy, and emotional support applications. The review also identified several studies that employ non-standard assessment techniques and some questions about transparency and the reproducibility of the studies [7].

Although LLMs are capable, they still have significant drawbacks in the medical field. Their answers can include incorrect or unverifiable details, as their information may be outdated or factually wrong, or because language models can produce plausible but inaccurate information, which is also known as hallucination. Besides this, there can be no clarity in the logic of generated responses. The shortcomings of RAG applications in healthcare identified in a 2025 systematic review are outdated training knowledge, hallucination, limited transparency, and a lack of standardized evaluation [8].

To enhance the grounding of the responses generated by the LLM, Retrieval-Augmented Generation (RAG) has therefore arisen as a strategy to retrieve relevant information from external knowledge sources and then feed this into the language model when generating a response. The healthcare literature on RAG, however, is disparate, and new studies in this field have found that there are no any standardized evaluation frameworks and that ethical issues are not covered in many existing studies of healthcare RAG.

The conventional RAG systems primarily use semantic or lexical similarity to retrieve relevant textual passages. This method can be useful for providing evidence for certain mental health questions but might fall short for more complicated questions involving understanding of relationships between symptoms, disorders, treatments, or comorbid conditions. Multiple symptoms may be reported together by the same patient, and the extraction of distinct pieces of information may not necessarily reflect the associations between the clinical concepts.

Such relationships can be explicitly represented using knowledge graphs. A Knowledge Graph can be used to model the clinical concepts as nodes and relationships between them so that there are several related concepts that a system can follow in the retrieval and reasoning. Multi-source retrieval and Biomedical Knowledge Graphs are also combined to tackle the issue of hallucinations and factual grounding in recent research in the field of healthcare. For instance, MEGA-RAG uses dense retrieval, BM25 retrieval, biomedical Knowledge Graphs, reranking and evidence-guided refinement and yielded over 40% fewer hallucinations in a comparison with evaluated baselines [9].

Considering these difficulties, HealLM aims to build a mental-health RAG system

addressing mental health issues, which is different from traditional document retrieval. The proposed system is based on Dynamic Bridge-Symptom GraphRAG architecture with a focus on the anxiety-depression comorbidity. It includes a knowledge graph of disorders symptoms, treatments, clinical outcomes, and more that includes evidence, provenance, confidence, timestamps, and graph version.

The project specifically embraces Anxiety–Depression Comorbidity as a clinical construct and links it with relevant disorders and symptoms. The system can then find overlapping/bridge symptoms and do graph-guided retrieval and reasoning.

To tackle these challenges, HealLM proposes to address the limitations of traditional LLM and RAG-based mental-health applications by integrating conversational mental-health knowledge, dynamic knowledge-graph representation, multi-hop clinical reasoning, evidence-aware retrieval, and safety-orientated response generation.

1.2 Motivation

HealLM aims to address the need for easily accessible mental health resources and the computational constraints of current AI-driven healthcare systems.

The extent of mental-health conditions worldwide and the ongoing mental-health treatment gap illustrate the need for more readily available sources of mental-health information. While AI systems do not act as a substitute for trained healthcare professionals, a well-designed AI assistant can help patients navigate the healthcare system by delivering patient-friendly health information, recognizing issues, and prompting follow-up care with a healthcare provider. The need is especially high in Bangladesh, where most people with mental-health problems do not receive the available mental-health services [10].

While LLMs are also excellent NLP tools, their application to healthcare has raised questions about their factual accuracy, hallucination, transparency, and safety [7]. Such restrictions can be especially significant for mental-health-related applications, as they could result in giving a wrong or inappropriate answer to a user, or not picking up on a situation where a professional response is needed. A recent study of LLMs in mental health has highlighted the lack of consistency in evaluation methods and issues of transparency and reproducibility.

RAG offers a computational approach to grounding the responses of LLM's with external knowledge. The field of healthcare RAG research, however, is still quite diverse with a wide variety of retrieval methods, data sources, evaluation approaches and ethical considerations. One recent systematic review identified a lack of standardized evaluation frameworks as an important limitation [8].

Mental health questions might require complex reasoning, such as relationships between the document and the question, rather than just a simple document match. Some concepts of clinical relevance can be linked to one another in a user's description, and some disorders can have similar symptoms. So, an AI system designed to assist with mental health issues must be able to correlate symptoms with disorders and detect potential for comorbidity does not treat each symptom separately.

Finally, there is a computational argument for fusing unstructured textual evidence and structured medical knowledge. To overcome this, HealLM utilizes a dynamic

knowledge graph, along with retrieval and language-model generation. The graph maintains supporting evidence, provenance, confidence, graph version, and conflict information to be able to evaluate and trace retrieved knowledge.
So, there is a need for an AI system for mental health that can be evidence-based, be aware of comorbidities, be fluent in responding, be explainable, and be safe.

1.3 Objectives

The main goal of HealLM is to design the intelligent and evidence-based mental health assistance system that can handle complex mental health questions, especially those related to anxiety–depression comorbidity. The goal of this project is to integrate Retrieval-Augmented Generation and a dynamic clinical knowledge graph for structured reasoning, evidence-based responses, explainability, and comprehensive evaluation.
The Main objectives of HealLM are:

- To design a mental-health-oriented Retrieval-Augmented Generation model coupled with a dynamic knowledge graph that represents and retrieves relationships between anxiety, depression, symptoms, comorbid conditions, treatments, risk factors, and other clinically related concepts

- To detect anxiety–depression co-occurrence and apply multi-hop graph-based reasoning to construct a model of the relationships among common and connecting symptoms to do multi-hop reasoning to extract connected clinical evidence for complex mental-health queries

- To incorporate evidence from various sources, using evidence fusion and verification, and to present answers in a way that is comprehensible by presenting clinical concepts, relationships, supporting evidence, and reasoning information in the answer generation process

- To support multi-lingual mental-health interaction in Bengali and English by using translation layer to translate the input query from Bengali to English so it can be interacted with the retrieval and reasoning pipeline and output the answer in Bengali or English
- To assess the performance of HealLM in terms of effectiveness, dependability and response quality through a series of comparative analysis by means of methods like retrieval, semantics, clinical performance, safety-related metrics, and the RAG metrics specifically designed for the evaluations of responses.

In summary, these goals focus on designing and testing a dependable, understandable, and comorbidity-aware mental health support system that can offer evidence-based answers and tackle the drawbacks of traditional LLM and RAG techniques.

1.4 Methodology

The HealLM project uses a methodology based on Graph Retrieval-Augmented Generation (GRAG), which involves generating a local mental health knowledge base, semantic retrieval, forming a structured Knowledge Graph, and a medical language model to generate responses. The overall process is intended to give answers to requests for mental health information by searching through a pre-curated set of medical information and adding structured relationships to important mental health concepts.

In the first step, a list of relevant medical books, research articles, scientific papers, and other reliable medical sources about mental health is compiled to build the knowledge base of the project. Resources that address concepts of anxiety, depression, symptoms, risk factors, treatment, comorbidities, and other mental health issues. The gathered documents are sorted and refined to remove unwanted material and get them ready to be retrieved.

The medical records are then segmented into smaller units of meaningful text that can be efficiently retrieved. A numerical representation is generated by including a model called embedding for each chunk, and this enables the system to find semantically similar information instead of just a keyword match. The embeddings created are then combined with the associated text information to create the local mental health knowledge database that the retrieval module of HealLM employs.

In addition to the textual knowledge base, a mental health Knowledge Graph is also built based on the medical corpora. The Knowledge Graph is a collection of clinically relevant concepts and associations that describe the relationships between them. This graph-based representation enables relationship-aware retrieval and reasoning, especially for queries that require reasoning between interconnected mental health concepts.

Inference is used to process the user input and retrieve information in the local medical knowledge base and Knowledge Graph. Retrieved information is unstructured textual evidence and structured clinical relationships that are given to the generator. The complementary sources are used as background information for MedGemma 1.5 4B [11] to create a natural-language answer based on the retrieved medical evidence.

A Bengali English translation layer implemented on NLLB-200 distilled 600M [12] is embedded in the system for supporting the interaction in the Bengali language. For a Bengali query, the user submits a query which is then translated to English and then fed to the retrieval, Knowledge Graph and generation components. Once MedGemma 1.5 4B generates the response in English, the translation layer will translate the response back into Bengali to present to the user.

Finally, the system is evaluated using multiple evaluation categories. Current evaluation measures are classification-oriented (Accuracy, Precision, Recall, Specificity, F1-score) as well as generation and semantic oriented (ROUGE-1, ROUGE-2, ROUGE-L, METEOR, Cosine Similarity, Exact Match). There are 570 examples in the current test set.

The methodology includes a combination of pre-selected medical knowledge, semantic retrieval, structured clinical relationships, the generation of medical languages, and multilingual translation, all in one. This architecture allows HealLM to fetch relevant evidence from several knowledge representations to deliver more contextually relevant responses to complex mental health queries that are also clinically grounded.

1.5 Project Outcome

The main product of this project is the creation of HealLM, an intelligent mental health assistance system that will yield evidence-based answers for mental health-related queries in natural language. The system is designed to integrate the retrieval of medical knowledge, the use of structured clinical relationships, and the generation of the response from a language model under a single framework.

The project is anticipated to generate a Knowledge Graph for mental health, which will structure and categorize key clinical concepts and relationships, focusing on anxiety, depression, symptoms, risk factors, treatments, and related conditions. The system can also be used to simulate anxiety–depression comorbidity and detect commonalities and/or interdependencies among symptoms, which can be helpful for understanding complex mental health queries.

The other expected outcome is the combination of text-based retrieval and retrieval based on Knowledge Graph to obtain relevant medical evidence from multiple knowledge sources. The current goal of the system is to enable graph-based reasoning, integration of evidence in the generation, and the integration of supporting information for clinical concepts and explainable responses by associating generated answers with relevant clinical concepts and supporting information.

The project will also feature a multilingual mental-health conversational interface, allowing users to converse with the system in Bengali, along with English. The integration of the translation component with the retrieval/generation pipeline is designed to make evidence-based mental health information more available to the Bengali-speaking users.

Lastly, the project aims to offer a framework for systematic assessment and comparison of the proposed approach to conventional LLM and RAG-based methods. Overall, HealLM is designed to showcase the potential of integrating medical knowledge retrieval, Knowledge Graph-based reasoning, evidence-grounded generation, and multilingual interaction to create a more structured, explainable, and complex approach to mental health question answering that is aware of the presence of comorbidities.

1.6 Organization of the Report

This report is divided into six chapters, which contain the background and research motivation for HealLM, the methodology, implementation, evaluation and conclusions of this project. The background of mental-health problems, their focus (anxiety and depression), motivation for the development of HealLM, project objectives, methodology for the project, and the expected project outcomes are presented in Chapter 1, Introduction. It also raises the importance of developing a mental health support system based on evidence and considering comorbidities.

Fundamental concepts and existing research on the project are discussed in Chapter 2, Background. It looks at the use of Artificial Intelligence and Large Language Models in mental healthcare, Medical Retrieval-Augmented Generation (RAG), clinical decision-support systems, Knowledge Graphs and GraphRAG, and multilingual medical AI. This chapter also includes the literature review and gap analysis which sets the motivation and positioning of the proposed HealLM system.

Chapter 3, Research Methodology, outlines the entire research methodology and system design of HealLM. It presents the system requirements, proposed architecture, data flow, user interface, local mental-health knowledge base, Clinical Knowledge Graph, intent classification, multi-hop graph retrieval, hybrid evidence retrieval, evidence fusion and validation, MedGemma 1.5 4B fine-tuning, response generation, safety verification, structured answer generation, and the NLLB-200 translation architecture. The chapter also summarizes the proposed algorithm, overall system architecture, project plan, and task allocation.

Implementation and Results are discussed in Chapter 4. It details the software and hardware configuration, HealLM GraphRAG configuration, MedGemma fine tuning setting, and evaluation environment. The chapter then presents information about the testing dataset, evaluation metrics, testing procedures, functional testing, performance analysis, and comparative analysis with MedGemma 1.5 4B. The findings are discussed for response generation, semantic similarity, semantic-match classification, anxiety–depression comorbidities and bridge-symptom reasoning, safety and multilingual response performance.

Chapter 5, Engineering Standards and Design Challenges, discusses the engineering aspects of the HealLM project. It encompasses aspects of adherence to relevant software, hardware, communication standards, project impact on society, environment, sustainability and ethics. In this chapter, project management and financial analysis are also introduced and HealLM is also mapped as a complex engineering problem using complex problem-solving, knowledge-profile and engineering-activity mapping.

The HealLM project's overall work and results are summarized in Chapter 6, Conclusion. It provides an insight into the significant contribution of the proposed system, its limitations, and some directions to further improvement and research. Finally, the chapter looks at the potential of all these elements in mental-health Q&A: evidence retrieval, Clinical Knowledge Graph reasoning, medical language generation, safety mechanisms, and multilingual interaction.

Chapter 2

Background

This chapter provides background and existing research as it relates to the work for the development of HealLM. It covers the basics of AI and Large Language Models in mental health, medical RAG, clinical decision support, Knowledge Graph, GraphRAG and multilingual medical AI, and a comparative study of the existing research and the research gaps that are identified by the proposed system.

2.1 Introduction

Mental healthcare is an integral component of healthcare in which timely access to reliable information, early recognition of symptoms, and appropriate professional support can make a significant contribution to patient outcomes. Mental health information, however, is frequently complex, with symptoms from different disorders being similar, conditions occurring at the same time, and multiple clinical factors needing to be considered when interpreting a user's description. The combination of these attributes poses difficulties for traditional information-retrieval systems and portrays the potential of Artificial Intelligence (AI) and Natural Language Processing (NLP) applications to the mental-health information management domain.

In recent years, with the advancement of Machine Learning (ML), Deep Learning, and Natural Language Processing (NLP) techniques, computer systems can process and comprehend huge volumes of unstructured medical text. Of these technologies, Large Language Models (LLMs) have proven themselves to be excellent at comprehending natural-language queries and producing coherent responses. Medical question answering, information retrieval, clinical documentation, and patient-oriented communication are examples of areas where medical language models can be used in healthcare. LLMs, however, depend greatly on the knowledge they have gathered over the training process and can produce inaccurate, outdated, or unsupported information. In mental health applications, these restrictions are especially significant, as they could hinder a user's ability to comprehend their well-being or postpone timely clinical intervention. Retrieval-Augmented Generation (RAG) is one way to tackle some of these challenges. RAG integrates information retrieval and language-model generation, retrieval of relevant information from external knowledge sources, and the subsequent provision of that information as context to the LLM. The system does not rely solely on the model's parameters but can be supplemented with a carefully curated library of medical resources to back up the answers with relevant evidence. The process of a typical RAG pipeline includes document collection, document preprocessing, text chunking, embedding generation, and retrieval based on vectors, as well as the generation of a response. In the domain of healthcare, this is beneficial, as medical information can be kept in sync with the language model without being dependent on it.

One of the extensions of RAG is Knowledge Graphs (KGs) to structure information. A Knowledge Graph is a network of nodes and edges that represent entities and their relationships, which is different from the traditional, text-based approach to retrieval. In mental health, entities can be disorders, symptoms, risk factors, treatments, and comorbidities, and relationships can be associations between disorders, symptoms, risk factors, treatments, or comorbidities. These types of structured representations can be beneficial for queries that involve comprehending the relationships among multiple clinical concepts, as opposed to retrieving a single piece of information.

The combination of Knowledge Graphs and RAG has led to the development of Graph-based Retrieval-Augmented Generation (GraphRAG) approaches. GraphRAG can leverage graph relationships to perform multi-step or multi-hop retrieval and find connected evidence. This is especially important in the context of mental health question answering, as anxiety and depression can have some common symptoms and can occur concurrently. It is important to recognize the relationships among such clinical evidence and to not consider them individually.

Multilingual accessibility is another critical factor when it comes to healthcare AI. Lots of medical AI systems and resources are built with English language data, potentially making them inaccessible to those who primarily communicate in other languages. A translation-based architecture can present Bengali queries in an interface that is translated into English to be retrieved, and English-language responses are translated back to Bengali for presentation to the user for users who speak Bengali. This architecture enables the system to be able to use available medical resources and to provide interaction in the user's preferred language.

Thus, building a mental health support system will be needed to combine multiple complementary technologies, not just LLM. Medical knowledge retrieval systems offer evidence from outside resources, Knowledge Graphs offer structured medical relationships, GraphRAG offers a process for retrieval and reasoning that is aware of relationships, and LLMs offer natural-language response generation. These components, coupled with multilingual translations and suitable evaluation systems, lay the technical underpinning for the development of HealLM, a proposed RAG system for mental-health-related applications. These components, along with multilingual translation and suitable evaluation systems, serve as the technical bedrock for the development of HealLM, the proposed mental health-related RAG system.

This background sets the baseline concepts needed to comprehend the following discussion of related research. Existing research on mental health AI, medical LLMs, RAG, Knowledge Graphs, GraphRAG, and multilingual healthcare systems is discussed in the next section, while existing research gaps are highlighted in the context of the proposed HealLM framework.

2.2 Literature Review

We can classify the existing literature on HealLM into five categories: AI and LLMs for mental healthcare; Medical RAG and clinical decision support; Knowledge Graph and

GraphRAG applications to healthcare and mental health; Low-resource language translation and graph-based approaches; Medical RAG and multilingual translation.

Limbic Access, an AI-powered decision support chatbot, was established by Mirecea et al. (2026) [13] for enhancing clinical preparedness and inhibiting burnout among mental health assessments. In a survey of 131 psychological wellbeing practitioners from nine NHS Talking Therapies services in the UK, AI-generated clinical summaries were compared with traditional referral information. The system is integrated with electronic health records. By using adaptive conversational intake to gather patients exhibiting problems, weigh risks, and structure patient information before consultations. The system results in higher clinician well-being and task performance on AI-supported assessments with lower cognitive load. Also, it benefits most pronounced among clinicians managing heavier caseloads. These outcomes are potentially scalable in AI decision support to reduce workforce burnout among mental health services. Nonetheless, the study relied on self-reported outcomes, emphasizing the need for randomized or eventual validation across a broader clinical population.

Sharma et al. (2026) [14] build a psychiatry-specific framework for agentic AI, lying between static decision support tools and fully autonomous systems. Using PRISMA-guided reviews of 28 studies, the authors offered a classification of assistive, collaborative, and semi-autonomous agents. These support a wide range of clinical activities, including medical documentation, multimodal symptom monitoring, crisis identification, different diagnoses, and personalized treatment under human supervision. Their work highlights a phased implementation strategy roadmap from low-risk pilot studies, followed by multi-site clinical trials, and concludes with long-term model performance monitoring. They also introduced a risk mitigation framework for addressing algorithmic bias, automation bias, privacy concerns, and regulatory uncertainty. According to the authors, Agentic AI has the potential to improve access to mental health care and plummet the workload of clinicians without trading the therapeutic correlation if human administration remains the cardinal of the system. The proposed framework is not quantitatively validated, since it is a theoretical perspective rather than an empirical study.

Qian et al. (2025) [15] discovered the aptitude of Foundational Models (FMs) to refurbish digital mental healthcare. The authors describe how large, pre-trained transformer-based models can integrate information from diverse data types, including text, speech, image, wearable sensor data, EEG signals, and social media, to support emotion recognition, early detection of depression and anxiety, personalized treatment, and constant mental health support. The study introduced a conceptual framework to map FMs through the full mental healthcare process, like data collection, prediction, intervention, and feedback. In addition, the authors discuss pairing FMs with brain-inspired models while maintaining clinicians in the loop to improve diagnostic accuracy and individual treatment. Although this framework provides a conceptual perspective rather than an empirical study, it offers no quantitative results or benchmark testing. The authors also identified some challenges like data privacy, algorithmic bias,

explainability, and clinical accountability.

Ge et al. (2025) [16] gave a detailed overview of the detection of mental health disorders using social media from deep learning to large language models (LLMs), to Retrieval-Augmented Generation (RAG), and to agentic AI systems. They categorized mental disorders into three categories (internalizing, psychotic, and externalizing) and reviewed techniques, information, assessment techniques, and models used for each disorder category. They also conducted experiments to compare the accuracy, contextual understanding, and explainability of different LLMs for a mental health detection task as well as measure the effect of RAG. The results showed that RAGs with trusted clinical knowledge sources can reduce hallucination, while agentic systems, which can reason, remember, and plan their multi-step interactions, can provide more personalized and autonomous support. However, the survey revealed several areas for potential improvement: noisy social media data, limited clinical testing and testing in the real world, small and inconsistent datasets, and privacy and ethical concerns. The authors recommended that future studies should focus on developing safe, effectively explainable, and clinically validated AI tools that would help ensure humans maintain control over mental health diagnosis and treatment procedures for safety and security purposes.

Mental health and substance use disorders are often co-occurring, and there are a handful of digital tools that focus on mental health and substance use separately. Agarwal and Ransingh (2026) [17] developed IntegraMind, a graph-based chatbot framework, with the aim of testing the feasibility of developing a unified system to address both mental health and substance use. They developed a multi-agent system in LangGraph that directly integrates four validated screening tools (PHQ-9, GAD-7, DAST-10, and CAGE) into conversation, a RAG module to retrieve evidence-based therapeutic content, and a human-in-the-loop safety layer that notifies therapists when crisis indicators are detected automatically. The system was tested using 492 synthetic test cases covering screening, crisis detection, and more complex multi-domain situations. The overall screening accuracy of IntegraMind was 96.88%, the crisis detection recall was 93.33% with 97.22% precision, and it achieved 100% accuracy in providing therapist alerts, well above the 45–78% crisis detection accuracy rates that have been reported for mental health chatbots. Despite this, the authors concluded that an evaluation was conducted entirely with synthetic, not real, patient information and emphasized the need for further research in the areas of real-world validation, cultural and linguistic adaptation, and longitudinal outcome studies before such systems can be implemented in clinical care.

Zhao et al. (2026) [18] introduced MedRAG, a RAG framework for healthcare copilots enhanced by knowledge graphs. Instead of typical RAG systems that use simple heuristic retrieval, MedRAG is built on a four-tier diagnostic knowledge graph from EHR data benefiting disease clustering, hierarchical accumulation, and LLM-based augmentation. Patient records to reason through similar-looking diseases, recommendation treatments, and generate follow-up questions are combined with the graph. It was tested on the public DDXPlus dataset and a private chronic pain dataset (CPDD) from Tan Tock Seng Hospital. It outperformed several RAG models in terms of both accuracy and diagnostic

specificity. However, the system heavily relies on good-quality EHR data and still needs testing across more medical fields.

To determine if open-weight, large language models can be used to generate guideline-consistent treatment recommendations for bipolar disorder, the authors of this study, Mali and Zeng et al. (2025) [19], created a retrieval-augmented generation (RAG) chatbot. They developed a multi-step RAG pipeline with Llama 3.3 70B that categorizes user queries based on their mood state, followed by a hybrid dense and sparse search to retrieve relevant sections from the 2018 CANMAT and ISBD guidelines, and then generate responses. System performance was assessed on 21 clinical vignettes in complex cases, including age, comorbidities, and medication history, with six expert psychiatrists rating the responses from both the RAG system and the base model system independently, in a blinded manner. The RAG system achieved statistically significantly more results that were fully consistent with the guidelines (63.5% compared to 19% for the base model) and fewer major errors (7.9% versus 38.1% across all the raters). Notwithstanding, the authors also noted that retrieval failures and shortcomings in multi-step reasoning still caused some inaccuracies and that more complex retrieval techniques, newer LLMs, and real-world, multi-turn clinical contexts would need to be explored in greater detail before they are deployed in the real world.

Singhal et al. (2025) [20] introduced Med-PaLM 2, a more powerful LLM for improving healthcare expert question answering. It fuses the PaLM 2 foundation model with medical fine-tuning, referred to as ensemble refinement, and a novel "chain of retrieval" approach that supports answers with reliable medical sources. Med-PaLM 2 achieved an accuracy of 86.5% on MedQA, outperforming Med-PaLM by more than 19%, and outperforming its predecessor on MedMCQA, PubMedQA, and MMLU clinical datasets as well. Physician evaluation of generated answers showed that Med-PaLM 2 outperforms those written by physicians on eight out of nine measures of clinical quality, including accuracy, reasoning, and medical consensus. The model also demonstrated greater safety; the risk of harm was lower for both the adverse and real-world clinical questions. However, the authors stressed that Med-PaLM 2 should be used in conjunction with, and not in place of, human healthcare workers, noting that more testing and validation are needed before the system could be widely deployed in clinical settings.

Singhal et al. (2023) [21] created Med-PaLM, a Large Language Model (LLM) for the medical domain, to measure its ability to accurately incorporate and utilize medical domain knowledge in medical question-answering tasks. They also developed MultiMedQA, a benchmark combining seven datasets: MedQA, MedMCQA, PubMedQA, LiveQA, MedicationQA, MMLU clinical topics, and their new HealthSearchQA dataset, which "covers medical exams, research questions, and actual consumer health queries. Med-PaLM was developed on top of the 540-billion-parameter Flan-PaLM model, which was fine-tuned using instruction prompt tuning, designed to enhance factual accuracy, reasoning, and safety. The model achieved a new state-of-the-art accuracy of 67.6% on MedQA, while clinician evaluation found that the model gave answers that were more scientifically accurate, avoided harmful answers, and were less biased than the base model. Despite these advantages, the authors determined that Med-PaLM is still not

comparable to human clinicians in several critical aspects and suggested it be further developed to base its response on trusted sources, express uncertainty, provide multilingual support, and enhance safety before being applied in real clinical contexts.

Hasan et al. (2026) [22] presented a culturally grounded framework that combines Retrieval-Augmented Generation (RAG) with a clinically validated knowledge graph (KG) to emphasize LLM-generated counseling support for para-counselors in Bangladesh. Using 402 Bangla counseling records, the author built a dataset. In which 69 cases were manually annotated with a multidisciplinary team. They constructed a KG, which has 308 nodes and 642 relationships, determining causal links among stressors, interventions, and outcomes. This framework was evaluated using several LLMs, including Gemini, GPT-4.1, Llama-3.3-70B, DeepSeek, and Gemma variants. Overall responses by KG support, BertScore F1, and human-rated clinical relevance achieved consistent outperformance. On the other hand, SBERT similarity shows a less consistent result because of culturally enriched responses that are often distinct lexically from reference text. However, the study was limited by an expert-curated dataset, which may not represent all real-world cases, and KG is restricted by available observations. In addition, the framework showed weaker performance on environmental and contextual analysis and recommended expanding the dataset and improving contextual modeling in future work.

Li and Qin (2026) [23] introduced DualGraphRAG, which is a dual-view graph-enhanced RAG framework. It improves multi-hop question answering through combining knowledge graph (KG) construction, query enhancement, and dual view retrieval. To build a knowledge graph (KG), it extracts triples from unstructured text using LLM and enriches the quality of queries with explicit and implicit nodes. It also retrieves both one-hop triples and shortest paths for guiding answer generation. The framework was evaluated on Natural Questions, HotpotQA, and 2WiKiMultihopQA, where it achieved comparatively more accuracy next to NaiveRAG, GraphRAG, and LightRAG, and was computationally 11.5x more efficient than GraphRAG. Yet, the system relies on automatic LLM-based triple extraction that may generate noisy knowledge. Also, the authors recommend that re-ranking of node- or domain-specific ontologies would need to be extended in specialized applications.

Gao et al. (2025) [24] presented the Mental Disorders Knowledge Graph (MDKG), which is a large language model (LLM)-based approach to create a comprehensive and contextually enriched knowledge graph in mental health research. MDKG merges information from PubMed literature and curated databases while retaining critical context such as demographic information, conditional statements, source information, and confidence scores, unlike conventional biomedical knowledge graphs. The result is a graph with more than 10 million relationships, with almost 1 million new relationships that aren't listed anywhere else. The authors demonstrated that MDKG significantly enhances prediction of major depression disorder, anxiety disorders, and bipolar disorder with UK Biobank data, and that adding this context reduces expert evaluation time by approximately 70%. MDKG is much broader than the current biomedical knowledge graph in the knowledge domain of mental health. The framework, however, requires continuous monitoring and additional clinical trials to validate its effectiveness and

reliability in healthcare environments.

Kumar and Ahmad (2026) [25] suggested a multilingual Public Health AI assistant with a Retrieval-Augmented Generation approach for delivering healthcare information in multiple languages. The system combines information about healthcare from various sources like the WHO and the Indian Ministry of Health and Family Welfare with a dense retrieval mechanism and an LLM-based generation layer. The framework uses NLLB-200-Distilled-600M to facilitate translation between over 200 languages in the process of multilingual interaction. The system obtained an end-to-end response time of around 1.8 seconds and a retrieval latency of less than 20 ms and successfully decreased the hallucination rate from 38.7% to 9.1%. The work illustrates that multilingual translation combined with RAG is possible for health care systems. The system, however, is geared more towards general public-health information and not specialized mental-health diagnosis or mental-health anxiety–depression comorbidity, which presents an opportunity for domain-specific medical RAG systems with more sophisticated knowledge retrieval and reasoning mechanisms.

Ghet, Oiu, and Nisioi (2025) [26] proposed an extremely low-resource language setting Graph-based Retrieval-Augmented Generation (RAG) for Aromanian–Romanian translation. The texts, aligned with sentences, used in the study were around 3,000, and the Aromanian–Romanian dictionary of 28,647 words. Using their RAG, they build a graph of Aromanian words, Romanian segments, and translation relationships to be able to retrieve directly and indirectly related language information. The authors benchmarked graph-based RAG against the LoRA fine-tuning of LLaMA and RoLLaMA models and NLLB-200 Distilled-600M. The best performance was obtained by NLLB with BLEU scores of 5.62 when translating Aromanian to Romanian and 5.18 when translating Romanian to Aromanian. Fine-tuning based on LoRA was not performing very well, with RoLLaMA3-8B resulting in BLEU 1.61 and chrF 18.26. For commonly used texts, RAG was promising, but it had issues with memorization and data contamination. From a general point of view, the study shows the ability of the graph-based retrieval method for low-resource translation, but it was not possible to get a reliable practical translation due to the very small training data set.

The reviewed studies highlight that there are significant advances in the application of AI, LLMs, RAG, Knowledge Graphs, and multilingual technologies to healthcare and mental-health applications. There are significant differences, however, in how they are aimed, what data they have, how they are retrieved, what kind of graph they have, how they are evaluated, and what population they are aimed at. The major studies that were used in this research are summarized in the following table.

Table 2.1: Summary of Literature Reviewed.

Author(s) & Years	Methodology	Key findings	Remarks
Mirecea et	Adaptive	Better clinician well-being,	Primarily uses self-

al. 2026 [13]	conversational intake, an AI-driven clinical decision-support chatbot with EHR integration, was evaluated by 131 psychological well-being practitioners	improved task performance, and reduced cognitive load, especially for clinicians with heavier caseloads	reported results; requires a wider clinical validation
Sharma et al. 2026 [14]	PRISMA framework to review 28 studies. Categorized psychiatric agents into assistive, collaborative, and semi-autonomous. Suggested a framework for risk mitigation	Agentic artificial intelligence can help with documentation, monitoring, crisis detection and diagnosis, and treatment, but this has to be supervised by humans and tested in the clinic.	A concept not backed up by quantitative experimental evidence.
Agarwal & Ransingh 2026 [17]	Multi-agent chatbot with integration of LangGraph for PHQ-9, GAD-7, DAST-10, CAGE, RAG, and human in the loop for detecting crisis.	In this evaluation, only synthetic cases were used, and screening accuracy, crisis recall, and precision were achieved at 96.88%, 93.33%, and 97.22%, respectively.	Needs to be validated in the real world, culturally.
Zhao et al. 2026 [18]	Healthcare RAG with a 4-level diagnostic KG based on EHR data; tested on DDXPlus and the chronic pain data set.	Performs better than traditional RAG systems in diagnostic accuracy and specificity but requires good-quality EHR data.	Highly dependent on high-quality EHR data
Hasan et al. 2026 [22]	RAG trained with a clinically validated KG made from 402 Bangla counseling records; evaluated using multiple LLMs and human evaluation.	Clinical relevance, as rated by humans, and F1 BERTScore scores improved with KG support, while dataset and contextual coverage scores were still low.	Weaker contextual coverage, limited data set

Li & Qin 2026 [23]	Dual-view GraphRAG, which integrates LLM-based KG construction, query enhancement, one-hop retrieval, and shortest-path retrieval.	Superior to NaiveRAG, GraphRAG, and LightRAG on multi-hop QA and 11.5× less time-consuming than GraphRAG.	Some noise may occur in KG due to LLM based, and a better domain specific ontology reranking is required.
Kumar & Ahmad 2026 [25]	Multilingual RAG with WHO/MoHFW Healthcare Sources, Dense Retrieval, LLM Generation, and NLLB-200-Distilled-600M for translation.	Conducted end-to-end latency of ~1.8 seconds and minimized hallucination from 38.7% to 9.1% but did not focus on specialized mental-health comorbidity.	Public Health Focused
Qian et al. 2025 [15]	Conceptual framework of the use of foundation models for multimodal mental health data, such as text, speech, images, EEG, and wearables data, as well as social media data.	Showed promise in detecting depression and anxiety and suggesting personalized interventions, but pointed to issues with privacy, bias, and explainability and accountability.	Conceptual (no quantitative benchmark evaluation)
Ge et al. 2025 [16]	A comparison and a review of deep learning, LLM, RAG, and agentic-AI methods for mental health detection.	RAG from trusted sources decreases hallucination, and agentic systems increase contextual and personalized reasoning, but there is limited real-world clinical validation.	Inconsistent dataset
Mali & Zeng et al. 2025 [19]	Hybrid dense/sparse retrieval and mood-state query classification from the CANMAT/ISBD guidelines, using Llama 3.3 70B.	The proportion of responses that were consistent guidelines went up to 63.5% compared with 19% for the base model, and the size of the major error group decreased from 38.1% to 7.9%.	Tested on a small test set

Singhal et al. 2025 [20]	Ensemble refinement and a chain of retrieval mechanism for evidence-supported medical QA in PaLM 2.	Successfully passed MedQA with a score of 86.5% and enhanced clinical accuracy, reasoning, and safety over Med-PaLM.	Additional testing is needed for deployment in the real world.
Gao et al. 2025 [24]	LLM-based KG with PubMed and curated databases, with demographics, conditions, sources, and confidence scores.	Produced more than 10 million relationships; further clinical validation is needed to improve prediction of depression, anxiety, and bipolar disorder.	The clinical validation needs to be further performed.
Ghet, Oiu & Nisioi 2025 [26]	Graph-based RAG with 3000 parallel sentences and a dictionary of 28,647 words and compared to NLLB-200 and LoRA-based LLaMA/RoLLaMA models.	NLLB got the most accurate results in translation, and GraphRAG had the potential to perform well with commonly occurring texts but experienced memorization and data-contamination problems.	Small parallel data set is used, but it is very small.
Singhal et al. 2023 [21]	Instruction Prompt Tuning for Medical LLM on Flan; tested with MultiMedQA benchmark.	Obtained an accuracy of 67.6% for MedQA and gains in scientific accuracy and safety, but not at par with human clinicians in important clinical aspects.	Performance remained below

Table 2.1 shows the relevant literature for this study. The studies reviewed illustrate the increasing use of LLM, RAG, Knowledge Graph-based retrieval, and multilingual AI in healthcare and mental-health applications. Works have demonstrated their success in various tasks, including clinical decision support, medical question answering, mental-health counselling, diagnostic reasoning, hallucination reduction, and multilingual access to healthcare information. The reviewed systems, however, tend to pay attention to separate aspects of these technologies and still suffer from the following challenges: clinical validation, retrieval reliability, multi-hop reasoning, contextual coverage, and domain-specific knowledge representation. Specifically, there has been little focus on explicitly modelling the anxiety–depression comorbidity and shared or bridge symptoms within a single mental-health RAG framework. The observations lead to the development of HealLM, the goal of which is to integrate the above-mentioned four elements to get

more comprehensive mental-health question answering.

2.3 Gap Analysis

The literature review presented in the previous section shows that there have been a variety of studies into different aspects of mental-health assistance with AI, medical RAG, Knowledge Graphs, graph-based reasoning and multilingual healthcare systems. These are spread amongst several approaches, and some key aspects of a comprehensive mental-health assistance system are underdeveloped. These gaps are identified systematically by comparing the main characteristics of the systems reviewed to the proposed HealLM framework as presented in Table 2.2. This comparison involves mental-health specialization, representation of anxiety–depression comorbidity, Knowledge Graph and graph-based retrieval, multi-hop reasoning, combined text–graph retrieval, explainable clinical reasoning, Bengali-language interaction, mental-health-specific knowledge bases, safety mechanisms, evidence information, evaluation, and clinical data integration.

Table 2.2 Gap Analysis in Existing Studies.

Features	**IntegraMind [17]**	**MedRAG [18]**	**Hasan et al. [22]**	**DualGraph-RAG [23]**	**Public Health AI Assistant [25]**	**HealLM**
Mental health focus	Yes	No	Yes	No	No	Yes
Anxiety-depression comorbidity focus	No	No	No	No	No	Yes
Shared/bridge symptom representation	No	No	No	No	No	Yes
Mental health Knowledge Graph	Yes	No	Yes	No	No	Yes
Graph based retrieval	Yes	Yes	Yes	Yes	No	Yes
Multi-hop reasoning	Yes	Yes	No	Yes	No	Yes
Combined text + graph retrieval	Yes	Yes	Yes	Yes	No	Yes
Explainable clinical reasoning	Yes	Yes	Yes	Yes	No	Yes
Bengali language interaction	No	No	Yes	No	No	Yes
Mental health specific knowledge base	Yes	No	Yes	No	No	Yes
Safety mechanism	Yes	Yes	No	No	No	Yes
evidence information	No	No	No	No	No	Yes

The comparison shows that there are some important capabilities provided in an existing system, but most of them are often handled individually. Examples of systems that use RAG and structured reasoning include IntegraMind and MedRAG; two examples of culturally grounded mental-health support are Hasan et al. and DualGraphRAG shows graph-based multi-hop retrieval. Public Health AI shows how it is possible to build an effective multilingual RAG and translate-based interaction between citizens and health authorities. But all the compared methods do not explicitly integrate the modelling of anxiety–depression comorbidity, shared/bridge symptom representation, mental-health-specific Knowledge Graph reasoning, textual graph-based retrieval, and Bengali language interaction in the same framework. The comparison also points out that HealLM is not trying to solve all the possible functions of a healthcare system. Specifically, the current system lacks clinical data integration in real time, meaning that it fails to integrate clinical data from live Electronic Health Records (EHRs), wearable devices, or patient information that is continually updated. This constraint is one factor that sets the work of HealLM apart from clinical data-integrated systems, and it could be a future path to realize the system. The overall gaps identified provide the motivation of the proposed HealLM framework. The primary goal of HealLM is to overcome the specific challenges highlighted in the literature, such as integrating medical knowledge relevant to mental health, modelling anxiety–depression comorbidity, and understanding relationships between shared symptoms, by incorporating the above features into a single system: Knowledge Graph-based retrieval and reasoning; Textual RAG; Evidence integration; Explainable responses; Bengali language interaction; Comprehensive evaluation. Therefore, the proposed work is not only in the direction of generating mental health responses, but also in the direction of enriching the question answering with structured clinical relationships and retrieved evidence to provide more contextually relevant and evidence-driven answers.

2.4 Summary

In this chapter, the background, current research, and research gaps for the development of HealLM are presented. It begins with the basics of AI and LLMs in mental health care, Retrieval-Augmented Generation (RAG), Knowledge Graphs, GraphRAG, and multilingual medical AI, laying the groundwork for the proposed system. Approaches that are related to mental healthcare, medical RAG and clinical decision support, Knowledge Graph and GraphRAG applications, and multilingual and low-resource language technologies employing AI and LLMs are reviewed. The review studies illustrate the potential of these technologies with respect to different clinical domains such as clinical decision support, medical question answering, mental health counselling, diagnostic reasoning, hallucination reduction, and multilingual access to health care. They also note that there are certain drawbacks to clinical validation, retrieval reliability, multi-hop reasoning, contextual coverage, and domain-specific knowledge representation. The gap analysis also identifies that these capabilities are mostly dealt with independently in the existing systems, and there is

very little work done on the following aspects of a unified framework: anxiety–depression comorbidity, shared or bridge symptom representation, mental-health-specific Knowledge Graph reasoning, and Bengali-language interaction. HealLM aims to seamlessly address these gaps by integrating textual RAG, Knowledge Graph-based retrieval and reasoning, evidence integration, explainable responses, Bengali-language interaction, and comprehensive evaluation. The current system does not include real-time data integration from clinical sources (area for future development). Overall, this chapter sets the foundation and motivation for the proposed HealLM framework. The design and implementation of the system are presented in the next chapters, followed by an evaluation and experimental results.

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Chapter 3

Research Methodology

This section introduces the research methodology that led to the development and investigation of HealLM and introduces design specifications needed for implementing the proposed system. The methodology is explained in terms of use of evidence-based retrieval and graph-based reasoning to address the research problem, and the design specification specifies the functional structure and operation of the implemented system.

3.1 Methodology

The section provides an overview of the methodology in HealLM's research and the design of the project. Important facets of its design include system explanation, suggested methodology, system requirements, data processing, and the layout of the user interface. This part outlines the functional and non-functional system specifications and creates the basis for building the system for mental health support.

3.1.1 Overview

HealLM aims to be an evidence-based mental health support system that combines Retrieval-Augmented Generation (RAG), a structured mental health Knowledge Graph, semantic retrieval, graph-based reasoning, a medical language model, and multilingual translation. The main goal of the system is to provide natural-language answers to mental-health queries with relevant information that is retrieved from carefully curated medical resources and that considers relationships between clinically relevant concepts. The methodology starts with sourcing medical books, research papers, scientific articles, and other credible mental health information on disorders and symptoms, risk factors, treatments, comorbidities, and related concepts. These resources are broken down and segmented into semantically structured text units to enable easier retrieval of relevant information. An embedding model is used to translate each chunk into a numerical vector representation so that information that is semantically similar can be identified, even if the words used in a user's query are not the same as those in the original source. The local medical knowledge base of the retrieval component is obtained from the resulting textual information and embeddings. The resulting textual information and embeddings are the local medical knowledge base that the retrieval component will use. This enables HealLM to draw upon an external knowledge source to augment evidence-informed response generation in lieu of relying solely on the internal knowledge of the language model.

HealLM utilizes a dedicated mental-health Knowledge Graph derived from mental-health-related medical corpora, in addition to the textual knowledge base. The Knowledge Graph contains elements and relationships that encapsulate important clinical concepts, such as relationships between anxiety, depression, symptoms, shared and bridge symptoms, risk factors, treatments, and other mental health-related concepts. This is a

structured representation that is used in conjunction with the text retrieval process and provides explicit relationships between clinical concepts. Inference: When a user poses a query, the query is processed and the relevant evidence is retrieved from the local medical knowledge base and Knowledge Graph. The text retrieval sub-module detects semantically relevant passages from medical texts, and the graph sub-module detects semantically relevant clinical entities and relationships. The Knowledge Graph also enables graph-based and multi-hop retrieval, enabling the system to trace relationships between clinical concepts. This is especially crucial for the project's emphasis on anxiety–depression relationships and comorbidity, in which a query can include symptoms or concepts related to both anxiety and depression. The retrieved information from these two sources constitutes complementary textual and structured evidence, which is then used for the generation of responses.

An evidence integration process is used to combine the textual and graph-based evidence retrieved and passed to the generator model of HealLM, MedGemma 1.5 4B [11]. MedGemma 1.5 4B takes the user's question. and the retrieved contextual data and returns a natural language response supported by evidence. The system thus uses detailed medical information from literature, as well as structured clinical relationships from the Knowledge Graph in the process of response generation. There's also a translation module that's seamlessly woven into the HealLM pipeline, enabling interaction in the Bengali language. If a user poses a query in Bengali, it can be translated into English before it goes into the retrieval and generation phases, and then the system can use the existing English medical resources and generation pipeline. The English language response may be translated to Bengali before presenting to the user after the response generation. This translation architecture allows for interaction in the Bengali language without having to maintain the medical knowledge base and generation pipeline entirely in Bengali. Hence, the processing flow is unified and integrates query processing, textual retrieval, graph retrieval, evidence integration, medical response generation, and translation.

Overall, HealLM combines three modules: medical knowledge retrieval, reasoning using KG, and MedGemma 1.5 4B generation, while also incorporating multilingual translation, delivering structured and evidence-based mental health responses. The methodology is the backbone of the system design, implementation, and evaluation of HealLM discussed in subsequent sections.

3.1.2 Proposed Methodology

The proposed methodology of HealLM integrates the principal components outlined above into a coherent framework for question-answering in mental health. The architecture makes sure the ingredients of local medical knowledge, semantic and lexical retrieval, reasoning over Clinical Knowledge Graph, evidence verification, generation of medical language models, checking for safety, providing a structured answer, and translating into Bengali are smoothly fitted into the core architecture of HealLM.

The process begins with a user submitting a mental problem query. In case the query was provided in Bengali, its translation into English will be facilitated by means of the built-in NLLB-200 translator thus enabling the system to process the incoming query.

After a query, both the Hybrid Evidence Retrieval modules carry out the analysis. In case of queries on comorbidity of anxiety and depression, the Clinical Knowledge Graph is used as well. The clinical entities are found and relations within the graph connecting anxiety, depression and the relevant clinical information are extracted through the multi-hop reasoning approach.

Moreover, text evidence and graph-based knowledge are transferred to the Evidence Fusion and Validation stage. The stage unifies the evidence reviewed and prepares suitable context for the response generator. Then, the structure context is combined with the query obtained for MedGemma 1.5 4B [11] to generate the answer.

The answer is later checked through Safety Verification, which estimates the response concerning the security and appropriateness of this mental health data. After this, the structured response is generated with the help of Structured Answer Generation to achieve better clarity of the message. In case the answer needs to be in Bengali, the previously structured response is translated back into Bengali.

Architecture keeps a clear separation between the two layers of knowledge and generation. The external knowledge is retrieved from the local medical database and Clinical Knowledge Graph, and the relevance of the knowledge is evaluated with the help of retrieval and reasoning elements. MedGemma's role is in language generation followed by safety, structuring, and translation of the generated output. The modularity of this architecture allows for independent evaluation of the corresponding components while preserving their integration into the system known as HealLM.

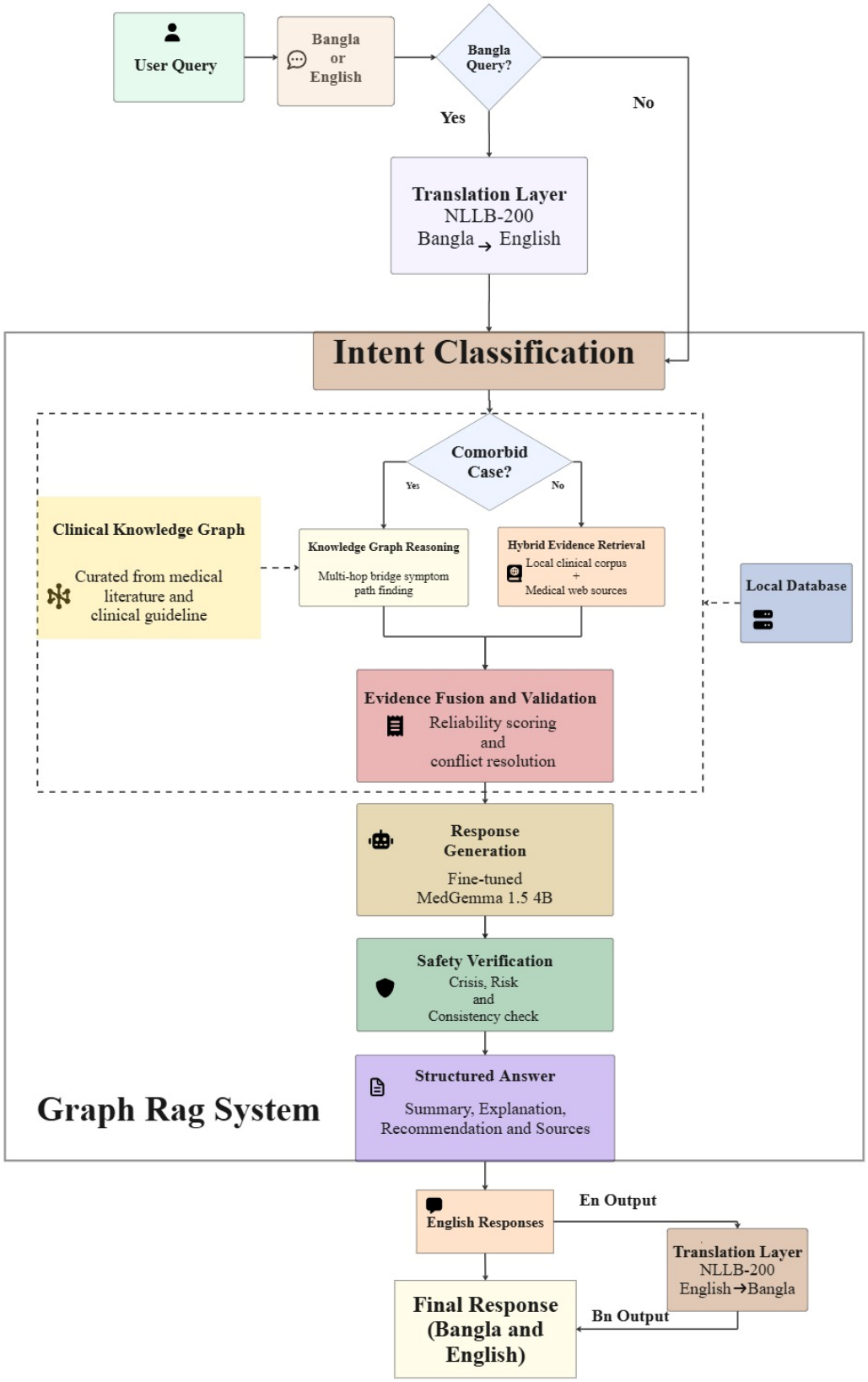


Figure 3.1: Proposed Methodology of HealLM

The set of procedures of the proposed HealLM GraphRAG system as illustrated in Fig 3.11 includes the steps starting from user query processing to language detection through the translation of the user's query from Bengali to English. The query is classified depending on the intent and the evidence retrieval procedure is carried out using the concept of knowledge graph reasoning where the cases of comorbidity are considered. The evidence that has been retrieved is then verified according to the results and response generation is carried out using a specialized application of MedGemma 1.5 4B, followed by checking for safety and answer generation. At the end, the appropriate response is either given in English or translated into Bengali, based on the preferences of the user.

3.1.3 Functional and Nonfunctional Requirements

3.1.3.1 Functional Requirements

HealLM requirements are broken down into functional requirements and nonfunctional requirements. Functional requirements are what the system must do, and nonfunctional requirements represent the anticipated qualities (or features), performance, reliability, usability, safety, and maintainability of the system. These requirements inform the design and development of major components of HealLM such as the medical knowledge base, Knowledge Graph, the retrieval modules, the MedGemma 1.5 4B [11]generator, the translation component, and the user interface.

Functional Requirements: With HealLM, users can enter mental health questions using natural language and obtain answers. The system can take queries in Bengali and English with a translation module to facilitate the exchange between the Bengali speakers and the English-oriented medical retrieval and generation module. The system receives a query and performs semantic retrieval in the local medical knowledge base to find the relevant information. The retrieval process involves computing embeddings from the processed medical resources and finding text passages semantically like the user's query.

HealLM retrieves information, besides textual retrieval, from a mental health Knowledge Graph that includes clinical concepts and relationships. This system detects relevant entities and relationships associated with the query and retrieves associated clinical information via graph-based retrieval. This facilitates multi-hop reasoning, in that relevant evidence can be gathered via many relationships on the graph. This functionality is especially critical for queries related to anxiety and depression, such as relationships, common symptoms, bridging symptoms, risk factors, and other related clinical concepts.

The system then conducts evidence integration by integrating and fusing the necessary information obtained from the textual knowledge base and Knowledge Graph. The integrated evidence is fed into MedGemma 1.5 4B to produce the final response in natural language. The response generation process is thus based on the query of the user and the medical information retrieved from outside sources. If Bengali output is desired,

the output in English is first translated into Bengali before it is provided to the user. The system is also designed to handle sensitive mental-health-related queries with safety-focused responses and is meant for information and support, not necessarily for a clinical diagnosis or for replacing professional healthcare.

3.1.3.2 Nonfunctional Requirements

HealLM will give exact, relevant, reliable, and evidence-based answers. The retrieved information should be relevant enough to the user's query, and the generated response should express the available medical evidence appropriately. The system should also have a reasonable response time to ensure that it can be used for retrieval, graph processing, evidence integration, evidence generation, and translation without making the conversation unfeasible.

In addition to this, usability and accessibility are important requirements of HealLM. The user interface must be simple enough to allow people without technical expertise to submit questions and comprehend the responses that are created. The support for the Bengali language should ensure meaningful interaction for users who speak Bengali but not block the use of the underlying English language retrieval and generation pipeline. The answer must be written in a clear and comprehensible way, especially about complicated mental health issues.

The system must be reliable and robust in processing a variety of mental-health queries and different natural-language formats. This modular design should also ensure maintainability and extensibility, meaning that the individual components of the system, like the retrieval mechanism, Knowledge Graph, embedding model, generator, or translation component, can be optimized or replaced without redesigning the entire system. This is also the opportunity to add more medical resources to the knowledge base in future versions.

Lastly, safety and privacy are some of the non-functional requirements of the system, due to its environment being the mental-health domain. It is important for HealLM not to provide generated content as a diagnosis or to imply that HealLM could be a substitute for a healthcare professional. The backlash to the questions should be a safe response. Information provided by users should also be managed responsibly, especially where a query includes information about a person's personal and/or sensitive mental health information. These are the functional and nonfunctional requirements that together define the operational and quality expectations that should be followed during the design, implementation, and evaluation of the HealLM system.

3.1.4 Data Flow Diagram

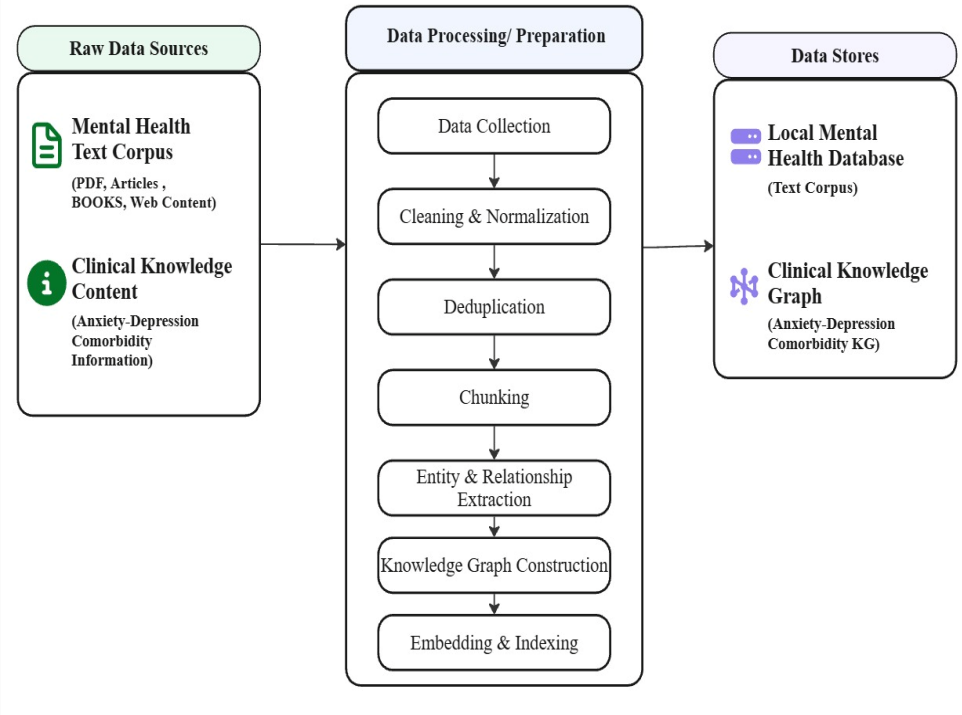


Figure 3.2: Data Flow Diagram

In the data processing and preparation workflow of the proposed HealLM system shown in Fig. 3.2, the raw mental health text and clinical knowledge undergo cleaning, normalization, deduplication, chunking, entity/relation extraction, and knowledge graph formation. Then, the processed data are embedded and indexed to form the Local Mental Health Database and Clinical Knowledge Graph, serving as data sources for the retrieval and reasoning steps later.

3.1.5 UI Design

The user interface design is presented in Fig. 3.3. In this figure, a user can put their mental health question into HealLM and receive the answer processed in a proper manner. The delivery system supplies all the necessary information concerning anxiety-depression comorbidity, the generated response, paths to comorbidity, and some links to local and other websites that can give further assistance.

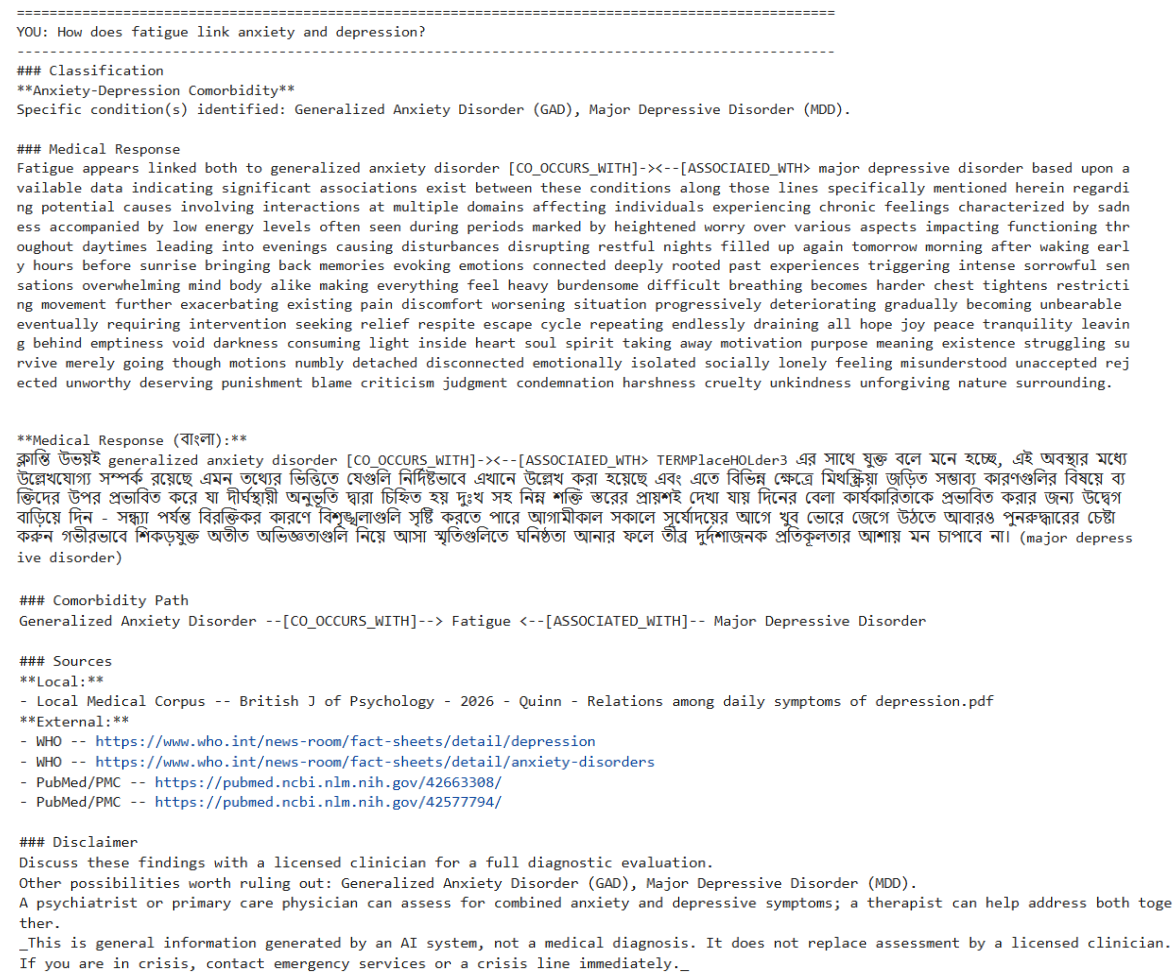


Figure 3.3: User Interface of HealLM

3.2 Detailed Methodology and Design

The detailed methodology and design of the proposed HealLM framework is given in this section. The design is based on medical knowledge retrieval, Clinical Knowledge Graph reasoning, evidence validation, medical language modeling, safety verification and multi-language interaction. Several alternative approaches are considered when designing, and the designed components get integrated accordingly based on their appropriateness to the mental-health question answering and focus on anxiety–depression comorbidity.

3.2.1 Local Mental Health Database Creation

The main textual knowledge source for the HealLM framework is the Local Mental-Health Database. 338 clinical resources, such as medical books, PDFs, research articles, journals, reference materials, and clinical resources for mental health are compiled and categorized to offer wide coverage of the concepts of mental health. The resources have been chosen to align with the key topics HealLM is addressing, namely depression,

anxiety, depression–anxiety comorbidity, screening and assessment, psychotherapy and treatment, neurobiology and genetics, and other clinically relevant topics. In Fig. 3.4 the distribution of the resources collected is shown with depression-related resources being the largest category (30%), followed by anxiety–depression comorbidity (15%), psychotherapy and treatment (14%), screening and assessment tools (13%), neurobiology/genetics (10%), anxiety (8%), guilt, shame and personality traits (7%), and reference textbooks and manuals (3%).

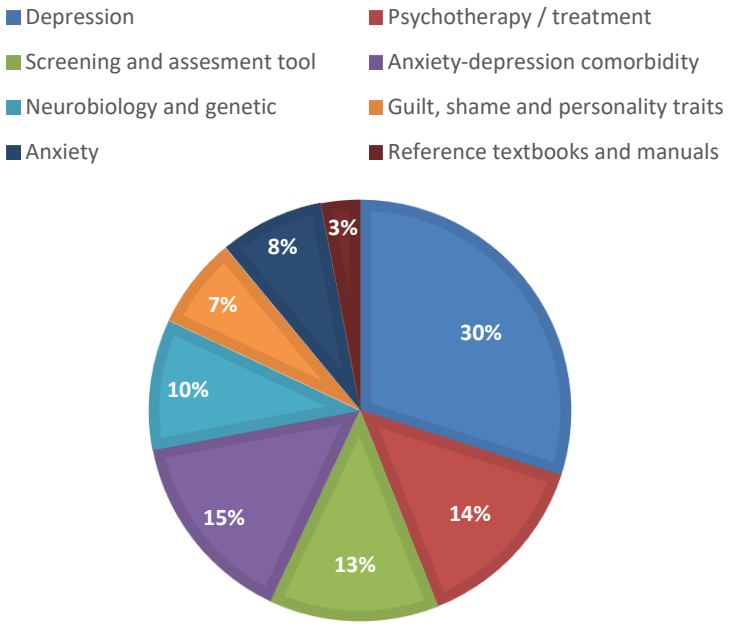


Figure 3.4: Mental Health Local Database in Percentage

Automatically collecting PDF resources from the configured local knowledge-based directory and processing them one by one. The backend package for PDF text extraction is primarily PyMuPDF (also known as fitz) which is efficient and well suited to scientific documents, and pdfplumber and pdfminer.six are available as fallbacks if needed. The text is then cleaned before being prepared for retrieval. Repeated headers and footers, standalone page numbers, DOI/ISSN lines, and too much whitespace are some of the common PDF-extraction artifacts that are removed by preprocessing. The system also detects poor quality components like table and figure captions, and fragments of citation lists, which would otherwise add noise to the corpus of retrieved elements. This pre-processing step ensures uniformity and quality of the textual information that is fed into the next retrieval stages.

The documents are then segmented into meaningful sentence-aware segments after preprocessing. HealLM has a target chunk size of about 300 words, and an overlap between chunks of two sentences. The chunking method does not chop text in the middle of words but attempts to keep sentences together in each chunk. It will also do chunking based on clinical sections, including Sections like Symptoms, Diagnosis, Treatment, Risk Factors, Medication, Prognosis, Side Effects, and Prevention. As a result, the chunks are intended to not join information from clinically distinct sections unnecessarily. When the length of a sentence is greater than normal it is split using a controlled word-based split

as a fallback. Very short trailing fragments and chunks containing low-quality extracted content are discarded.
The chunks are processed using Sentence-Transformers framework and mapped to dense vector representations with the embedding model BAAI/bge-large-en-v1.5 [27]. This is the default embedding model that is used in the implementation, while other embedding models are available if the default doesn't exist. The embeddings created are L2-norm normalized, so that cosine similarity can be computed efficiently as a dot product in the vector space when performing retrieval. The system also creates retrieval artifacts for both dense and sparse search, with the dense representations fed into the lexical retrieval stage in the following hybrid retrieval stage.

Therefore, the local database is a structured, searchable set of 338 curated mental health resources, and the pre-processing, clinical section-aware chunking, and embedding stages convert the original documents into a set of medical knowledge chunks ready to use in retrieval. This database is used as the textual evidence base for the following hybrid retrieval, evidence integration, and MedGemma 1.5 4B [11]-based response generation parts of HealLM.

3.2.2 Clinical Knowledge Graph

The structured knowledge part of HealLM is the Clinical Knowledge Graph (CKG), which consists of mental health concepts and their clinical relationships in a machine-readable graph structure. The graph has been built in an incremental fashion as more documents were added to it; the first building is based on 94 documents, and the second on 79 documents. The final graph was then supplemented with a clinical metadata annotation and a dedicated Anxiety–Depression Comorbidity (CAD) center node following the conflict detection. The graph contains evidence from the SeedCorpus, PubMed, WHO, NICE, NIMH, PMC and Wikipedia; and the relationships carry supporting evidence and provenance, confidence, weight and support-count information. For instance, the graph shows the relation between female sex and GAD and MDD as increases_risk_of and along with it, its evidence and its confidence value.

Several clinically relevant entity types are represented in the CKG which contains 142 nodes and 722 relationships. The disorders, the symptoms, the risk factors, the protective factors, the medications, the drug classes, the psychotherapies, the assessment scales, the clinical outcomes, the comorbid conditions, the neurotransmitters, the biological mechanisms, the genes, and the diagnostic criteria are represented as the main components of the graph with a specific Comorbidity Construct representing anxiety–depression comorbidity. The components enable the graph to depict mental health knowledge from different complementary clinical angles: disorders are the target conditions; symptoms are the observable manifestations of the disorders; risk and protective factors are a depiction of vulnerability and resilience; medications, drug classes, and psychotherapies are treatment options; assessment scales and diagnostic criteria are used during clinical assessment; biological mechanisms, neurotransmitter, genes, comorbid conditions, clinical outcomes, and patient-population nodes are additional

clinical information. Some of the entities in the canonical graph representation of the network are GAD, MDD, fatigue, insomnia, irritability, difficulty concentrating, panic disorder, social anxiety disorder, childhood trauma, chronic stress, family history of mental illness, female sex, social isolation.

These associations are clinically meaningful and reflect relationships between these entities, rather than isolated and disconnected pieces of information. For instance, the graph depicts symptoms by associated_with, risk factors by increases_risk_of, comorbid conditions by comorbid_with, treatments by relationships like treated_by and improves, assessments by relationships like measured_by, etc. For example, there is a relationship between social support and PHQ-9 measured_by, and physical exercise is represented as a protective factor that is associated with a lower risk and better symptoms of MDD. Likewise, the graph explicitly represents the cardiovascular disease as a condition which is comorbid with MDD, thereby providing a wider clinical context.

One of the key design features of the CKG is that GAD–MDD is explicitly modeled. GAD and MDD are shown as two target disorder nodes and the CAD center node gives an explicit structural representation of the comorbidity between GAD and MDD. GAD is linked to the CAD construct with the curated part_of relationship, signifying that it is linked with the core anxiety aspect of the construct of the comorbidity. This is especially relevant for HealLM as this allows the system to gain access to information per disorder but also information shared between anxiety and depression.

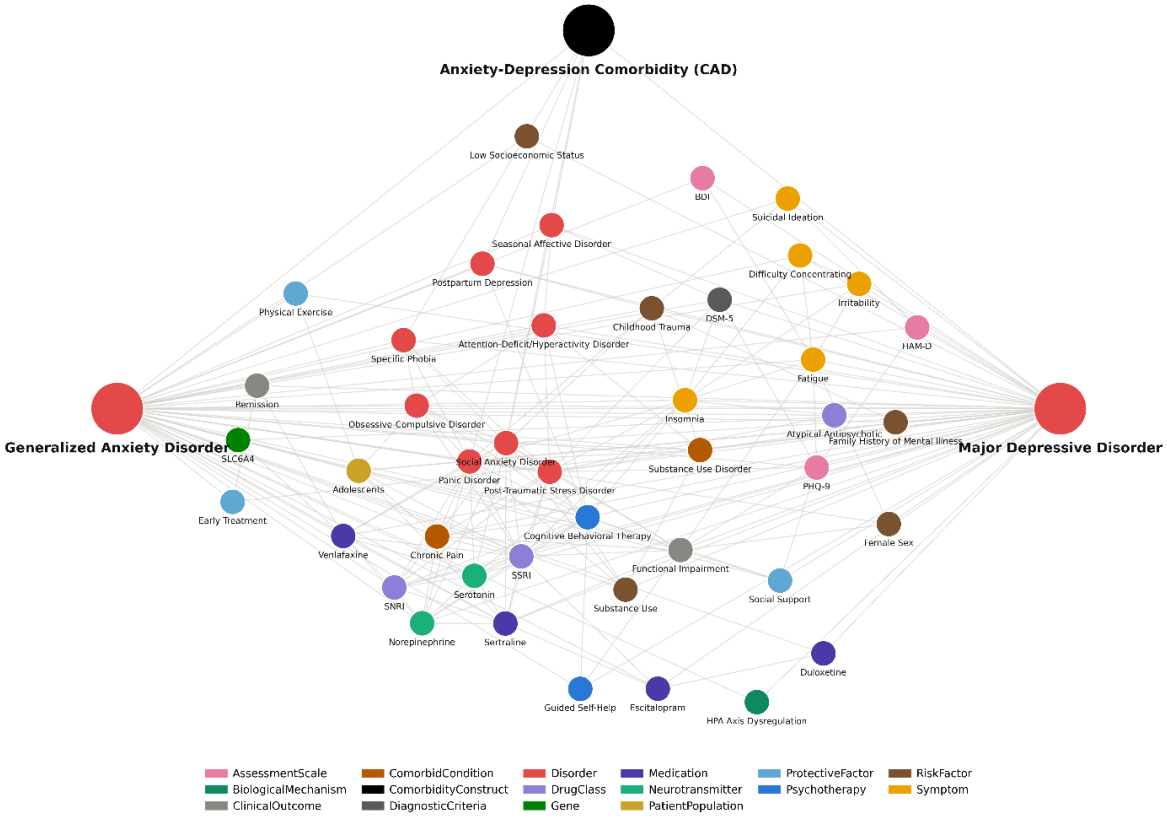


Figure 3.5: Knowledge Graph

This common neighborhood is shown on Fig. 3.5. The visualization reveals 44 nodes that are directly linked to both GAD and MDD, creating a wide linkage between the two target diseases. The disorders, symptoms, risk factors, medications, protective factors, drug classes, assessment scales, comorbid conditions, clinical outcomes, psychotherapies, neurotransmitters, biological mechanisms, genes, patient populations, diagnostic criteria, and the CAD construct are shared nodes. The graph depicts the GAD–MDD relationship for a few clinical dimensions, and not just shared symptoms of the two disorders.

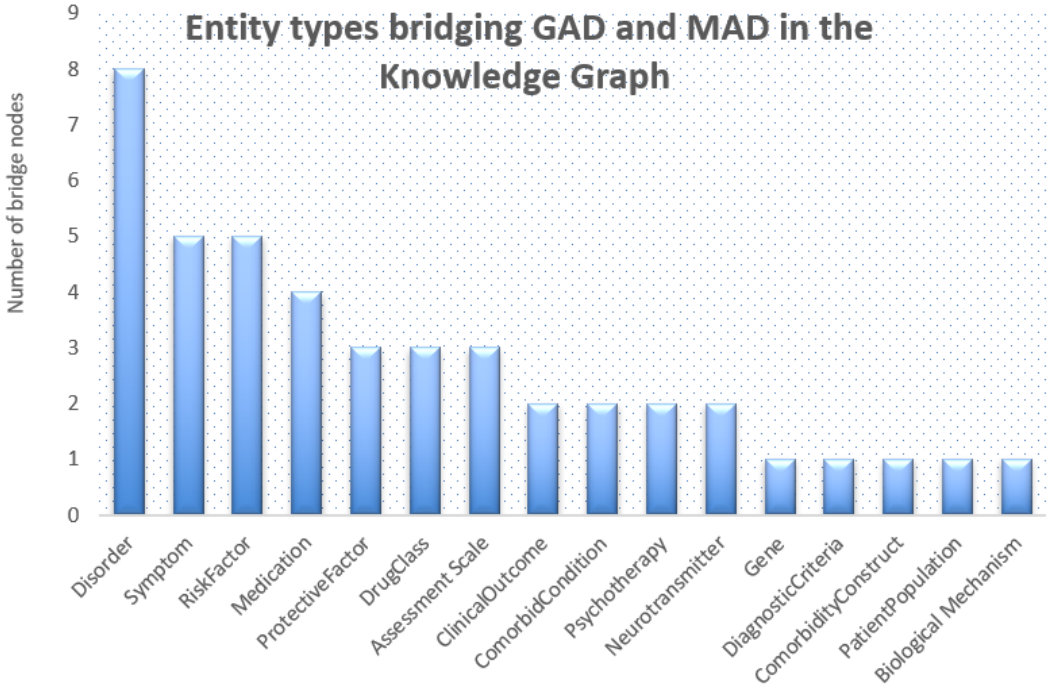


Figure 3.6: Bridge Nodes of Knowledge Graph

Fig. 3.6 shows these 44 bridge nodes that are broken down by entity type quantitatively. Most common are the nodes of the category Disorder (8 nodes) and Symptom and RiskFactor (5 nodes). Medication provides 4 nodes, and ProtectiveFactor, DrugClass and AssessmentScale each provide 3 nodes. Each of the following contribute 2 nodes: ClinicalOutcome, ComorbidCondition, Psychotherapy, Neurotransmitter, and each of the following contributes 1 node: BiologicalMechanism, Gene, DiagnosticCriteria, PatientPopulation, ComorbidityConstruct. This distribution shows that the common neighborhood of GAD–MDD is not uniform but rather is multi-layered with respect to clinical knowledge. We specifically highlighted common clinical features including fatigue, insomnia, irritability and difficulty concentrating; the graph explicitly shows the relationship between these overlapping symptoms in the context of GAD and MDD.

The bridge structure is important for the multi-hop reasoning capability of HealLM. One query about anxiety, depression and sleep disturbance, for instance, may then pass from the GAD and MDD nodes to the shared insomnia node and then on to appropriate treatments or assessment concepts. Similarly, a question for a shared risk factor can flow through shared risk factor nodes to either disorder. This relational relationship enables GraphRAG to return related clinical concepts instead of just the passages in the

documents that match a query's words.

In sum, the CKG is the clinical backbone for HealLM, which is relationship aware. The diverse clinical conditions, semantically rich relationships, evidence and provenance information, and explicit CAD-centric structure allow the system to store and retrieve the related knowledge of anxiety–depression comorbidity. Most crucial of all, the 44 shared nodes from Figures 3.5 and 3.6 show that the graph models the comorbidity as a multi-dimensional clinical network, thus establishing the structural framework for identifying the bridge nodes and multi-hop reasoning between GAD and MDD.

3.2.3 Intent Classification

HealLM uses the Intent Classification component to classify a user's mental health query prior to the retrieval and reasoning processes. This step is important because there may be different ways to retrieve the data, depending on what kind of query is asked. HealLM is particularly capable of recognizing queries related to anxiety–depression comorbidity from queries capable of being handled by the general textual and evidence-retrieval pathway. The classification is thus not a diagnostic model but forms part of the entire GraphRAG architecture as a routing mechanism.

When the user enters a query, the query goes through the language-processing phase first. In case of input in Bengali, the query is translated to English and then processed by the intent. The intent-classification phase then processes the semantic content of the query and categorizes it as either a comorbidity-related case or not. The knowledge graph reasoning component is used to handle queries that contain concepts or symptoms that suggest a relationship between anxiety and depression. Other mental health questions refer to a hybrid pathway of evidence retrieval. In this way, HealLM can skip graph-based multi-hop reasoning for queries that do not need relationship-based reasoning.

The classification is especially relevant for the primary research goal of the project to identify the anxiety–depression comorbidity and common symptoms. For instance, a query about anxiety with depression symptoms, or a query asking about both types of symptoms, may be a comorbidity-based query. The Knowledge Graph can then be used to enrich the query with relationships between related clinical concepts, like GAD and MDD, symptoms, risk factors, treatments, etc. The symptoms include common symptoms such as fatigue, insomnia, irritability, and difficulty concentrating, which can give further structure to these queries in the graph.

The system mainly adopts the Hybrid Evidence Retrieval pathway for non-comorbidity queries, which are basically used to retrieve the information corresponding to the query from the local medical knowledge base and accessible medical evidence. A simple informational query might not require multi-hop graph traversal, but a query pertaining to the relations between anxiety and depression can be useful with graph-based reasoning. Thus, intent classification will be utilized to decide on the right retrieval and

reasoning pathway while maintaining these two pathways within the same HealLM framework.

Another way would be to run graph reasoning on all the queries, no matter what their type is, as they are passed through the Knowledge Graph. This might give good graph coverage but would give irrelevant relationships to the user's query and process simple questions unnecessarily. A simpler alternative would be to use only textual semantic retrieval, without making an explicit distinction between queries that need some kind of relationship reasoning, which need to be related to the presence of a comorbidity. Therefore, to selectively enable the graph reasoning pathway where it is needed, HealLM employs intent-based routing.

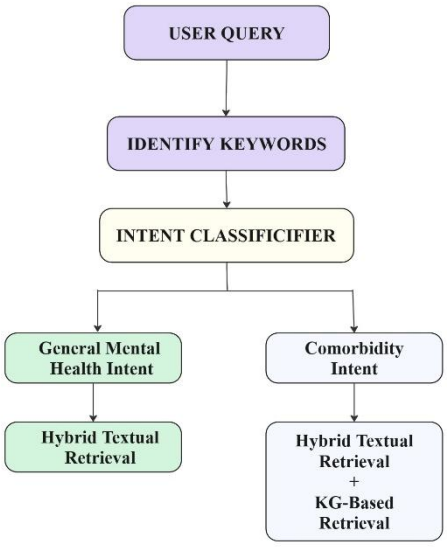


Figure 3.7: Intent Classification

Fig 3.7 illustrates how the HealLM system performs intent classification and retrieval. The user query is first classified into either the General Mental Health Intent, or the Comorbidity Intent. General queries are processed through hybrid textual retrieval, whereas comorbidity-related queries are processed through both hybrid textual retrieval and knowledge-graph-based retrieval.

3.2.4 Multi-hop Graph Retrieval

The Knowledge Graph Reasoning component is one of the primary research components of HealLM, which is mainly switched-on during queries classified as anxiety–depression comorbidity queries. It is used to identify significant relationships between clinical concepts beyond just retrieving isolated clinical concepts with the help of the structured Clinical Knowledge Graph. The reasoning process is especially crafted to identify bridge symptoms and multi-hop relationships between Generalized Anxiety Disorder (GAD) and Major Depressive Disorder (MDD). In the actual pipeline, graph reasoning is only used for the selected route by the intent classifier, which is the one that contains the comorbidity.

If a query is deemed to be a graph reasoning query, the system first resolves the clinical entities listed in the query with the actual Knowledge Graph. This is crucial since the reasoning process is query-dependent and not necessarily a fixed path GAD–MDD. The implementation is to solve the query's entities, and the entities matched to the graph nodes become reasoning anchor points. The graph paths for different queries therefore vary, so as a result, the exact path will vary depending on the clinical concepts found in the query.

If the disorder is anxiety–depression with reasoning, it will look for the clinically linked clusters of nodes between these two nodes and find intermediate clinical concepts linking them. There may be intermediate concepts shared across both that are symptoms, risk factors, treatments, assessment scales, biological mechanisms, or other clinically relevant entities that are represented in the graph. Of these, bridge symptoms are important to HealLM as they give an important clinical link between anxiety and depression. For instance, because of the commonality of symptoms associated with both disorders, insomnia, fatigue, irritability, and difficulty concentrating can be used to provide useful structural connections for retrieval oriented towards comorbidity. The reasoning process can be understood using a graph representation:

$$G = (V, E)$$

where V represents clinical entities and E represents relationships between clinical entities. A chain of reasoning can then be represented as a series of interrelated nodes:

$$v1 \rightarrow v2 \rightarrow v3 \rightarrow \cdots \rightarrow vn$$

The intermediate nodes are used to show the relationship between the two concepts (the starting and ending concept). In the case of HealLM, the resulting path is not considered a diagnosis. Rather, it gives enough structured evidence that reflects the relationships between the concepts in the underlying medical knowledge.

An important feature of the reasoning process implemented is that the confidence and supporting evidence of graph relationships are considered. The relationships include evidence and provenance information, and the reasoning part can differentiate between stronger and weaker supported relationships by using the evidence and provenance in the Knowledge Graph. The reasoning processing can be exposed as part of the structured response and be included in the final pipeline output, which includes Selected reasoning path, Graph nodes used, Evidence ranking information, Confidence information.

The reasoning component also considers multi-hop traversal, not just traversing nodes directly connected. Therefore, it is significant that clinically relevant data can be spread over multiple relationships. For instance, a question about anxiety and depression might need to be preceded by finding commonality in the symptom of anxiety or depression and then linking that symptom to either treatment or assessment. These interrelated concepts can then be retrieved as a reasoning chain with multi-hop traversal. The system thus focuses on the use of the graph as a tool for finding connected clinical evidence.

Another crucial design decision is that the Knowledge Graph is read-only at query-time reasoning. The current answer can be based on external or local evidence, but the evidence is not written in the graph in the process of inference. When the main answer() pipeline

is run, the graph is explicitly declared as a read-only graph. This separation ensures that individual user queries will not change the underlying clinical knowledge structure and maintain the reproducibility of graph reasoning for the same state of a graph.

The reasoning results then go to the evidence-integration stage, along with the textual and external evidence. The answer reveals the nodes of the graph as well as the path of clinical reasoning followed in the derivation of the final answer. For the response-structuring component, the reasoning is explicitly identified in the answers for the queries regarding the presence of comorbidity, but not for the ones regarding its absence. Direct graph retrieval (only nodes that are explicitly connected to the query nodes) is one important alternative that was considered. This method is less of a problem when there are no concepts in between relationships but can overlook relationships that may need intermediate concepts. The other way is to reason about the graph for each mental health query. This, however, results in needless graph processing for simple information requests. For this reason, HealLM employs intent-based graph activation, which activates only the multi-hop reasoning path for the queries concerning the comorbidities, and otherwise uses the hybrid evidence-retrieval pathway. In general, Knowledge Graph Reasoning enables HealLM with a relationship-aware reasoning capability in complex mental health queries. It involves query-specific entity resolution, multi-hop traversal, identification of bridges, evidence-aware path selection, and reporting the explicit reasoning paths, which are essential to the main goal of modelling the comorbidity of anxiety and depression. The system doesn't just compare the meaning of the query and medical passages but can also utilize the structure of the Clinical Knowledge Graph to discover related clinical evidence and present the related context to the response-generation step.

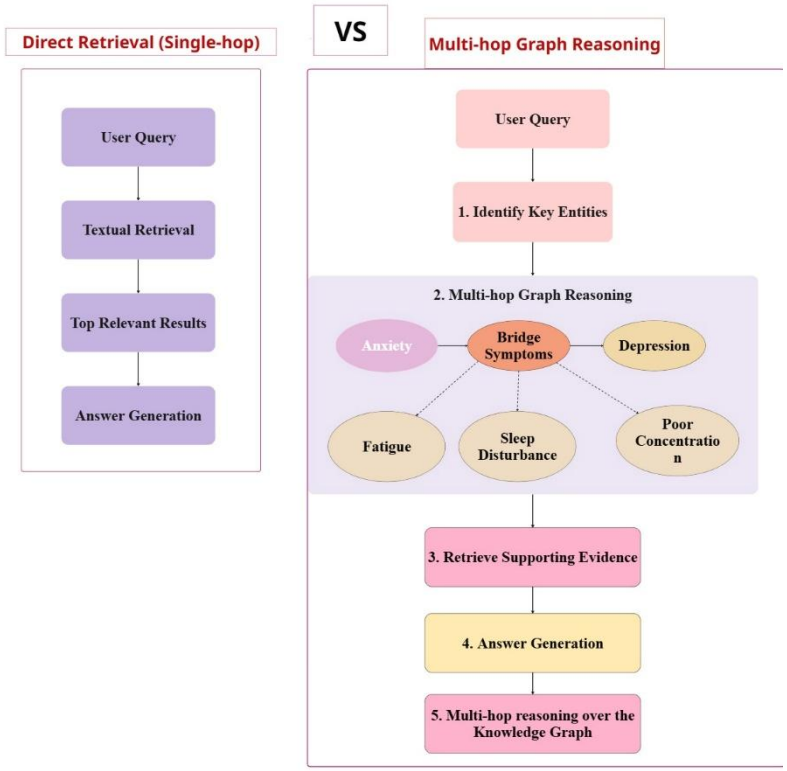


Figure 3.8: Direct Retrieval Vs Multi-Hop Graph Retrieval

As illustrated in Fig. 3.8, any user might notice the distinct difference between direct retrieval technique and our proposed multi-hop knowledge-graph reasoning approach. Direct retrieval technique provides the relevant text in one step whereas multi-hop technique identifies important entities, establishes relationship using bridge symptoms like fatigue, sleep disturbance, and poor concentration, retrieves evidence and uses this relationship for reasoning and answering in terms of comorbidity.

3.2.5 Hybrid Evidence Retrieval

The Hybrid Evidence Retrieval component is the one that handles the retrieval of medical evidence needed for the textual knowledge sources that are available to HealLM. The motivation of this component is to enhance the reliability of retrieval, not only relying on a single retrieval method (lexical sparse retrieval), but also using the semantic dense retrieval. This is especially helpful in the field of mental health, where a patient might use a different word for the same clinical concept, and for clinically important terms, disorder names, symptoms, medications, or assessment scales, where only exact lexical matching will help.

Dense semantic retrieval is used for the first retrieval pathway, based on the embeddings generated by the local mental health database. The documents in the local database are embedded according to the BAAI/bge-large-en-v1.5 [27] model, and the comparison between the semantic representation of a user query and the semantic representation of each stored medical text chunk is made. Dense retrieval can thus identify paragraphs that are conceptually related to the query even if the exact words used in the query are not found in the rest of the document. This is crucial for natural-language mental-health questions, as users could provide a symptom description using a different word order than that in medical literature.

The second way involves sparse lexical retrieval, where the query terms' occurrence and importance in the medical documents being indexed are emphasized. When the searcher knows specific clinical terms, the retrieval strategy is appropriate. Sparse retrieval may then be used alongside dense retrieval to retain the exact term that is not necessarily well captured via semantic similarity.

The information retrieved from these complementary methods will then be aggregated to create the hybrid evidence set. We don't consider dense retrieval and sparse retrieval as mutually exclusive strategies but rather as complementary sources of evidence, which is how HealLM treats them. Dense retrieval adds semantic matching, and sparse retrieval adds lexical relevance. The whole retrieval is then more representative of a range of candidate medical passages to be processed in the evidence. This design is consistent with the motivation found in the literature review, in which hybrid dense/sparse retrieval has been applied to enhance retrieval from clinically relevant information from the medical resources that are based on guidelines.

When no comorbidity query is used, hybrid textual retrieval is used as a main evidence-retrieval mechanism. A question on, for instance, depression symptoms, anxiety

management, psychotherapy, medication, or a screening instrument can be correlated with a local medical knowledge base to find relevant passages. Hybrid textual retrieval is used in combination with the Knowledge Graph reasoning component in queries related to comorbidities. In this case, detailed medical explanations are provided by textual evidence and multi-hop context, and structured relationships are provided by graph retrieval, enabling reasoning for anxiety–depression.

HealLM also uses external medical evidence for retrieval, if applicable. The system architecture specifies that the local medical corpus is one of the main evidence sources, while the information obtained from the medical web or other outside sources is used to supplement the information obtained from the local medical corpus. This gives another layer of evidence for where the local corpus is lacking in contextual coverage. The retrieved information can then be fed back to a component for evidence fusion and validation, which will evaluate the evidence sourced from the different components and feed it to the generator.

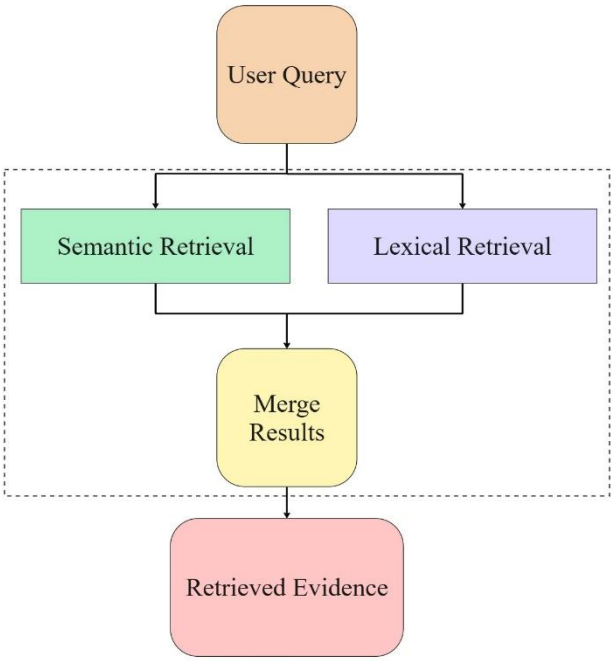


Figure: 3.9: Hybrid Evidence Retrieval Component

Fig 3.9 shows the components of hybrid evidence retrieval One such alternative that was considered is dense-only retrieval. While dense retrieval is good for semantic matching, it may not always be effective in prioritizing exact clinical terminology and extremely specific medical expressions. The second option is sparse-only retrieval, where lexical matching is good but semantically similar descriptions are not necessarily retrieved when the different terms are used. The third option is entirely to rely on the Knowledge Graph to retrieve. But a graph alone isn't medical literature: it does not provide detailed explanatory passages. Hence, HealLM adopts hybrid textual retrieval as its primary textual evidence retrieval method and, if necessary, adds textual evidence retrieval based on Knowledge Graph.

The retrieved evidence is not directly treated as the final answer. Rather, it goes into the Evidence Fusion and Validation stage, where the information from the various retrieval sources may be fused together and validated prior to the generation of responses. Such separation of retrieval and generation is key because the function of the retrieval component is to present the evidence relevant to its task, whereas the MedGemma 1.5 4B [11] generator must use that evidence to produce coherent natural-language output.

The overall Hybrid Evidence Retrieval component allows HealLM to have two evidence retrieval methods: semantic evidence and lexical evidence from the local mental-health knowledge base, and external medical evidence as an additional source if available. It is also integrated with Knowledge Graph retrieval, which allows the system to learn both extensive text information and structured clinical relationships that serve as the evidence basis for the subsequent validation, response generation, and safety verification.

3.2.6 Evidence Fusion and Validation

The Evidence Fusion and Validation component brings together the information from various knowledge sources of HealLM prior to feeding it to the response-generation model. The retrieved evidence includes mainly the local mental health knowledge base and the Clinical Knowledge Graph, and if it is applicable, external medical evidence is also added. The intent of this part is to give the generator a cohesive and pertinent evidence context, not based on a single retrieved passage or relationship.

The evidence when a question concerns general mental health is primarily drawn from relevant text passages retrieved in the hybrid fashion. In the case of queries associated with anxiety–depression comorbidity, textual information is supplemented by knowledge graph reasoning. The graph component is used to offer structured clinical relationships, shared concepts, and multi-hop reasoning paths, and the textual retrieval component is used to provide detailed explanations from the curated medical resources. These sources enable the system to take text plus structured clinical evidence into account when creating the response context.

The evidence retrieved is processed during the validation process, which considers relevance, consistency, source of information, and confidence in the retrieved evidence. The relationships within the Clinical Knowledge Graph are also stored with provenance and confidence information, allowing the system to hold information about the evidence supporting sources related to the relationships. This information is especially valuable if there is a reasoning path involving multiple relationships that would be part of a comorbidity. In the preparation of the final context, evidence that is more relevant to the query of the user is prioritized and is better supported by the available evidence.

The process of fusion also aids in eliminating unimportant or irrelevant information. All the retrieved evidence is not fed into the generator directly; rather, the evidence is put together in a coherent context. This context can include the medical passages supporting a query, as well as the graph relationships or reasoning paths that are relevant to it, in the case of a query about a comorbidity. This gives MedGemma 1.5 4B [11] a more

extensive clinical context and helps to link the response generated with the retrieved evidence.

Another way would be to directly feed the retrieved documents and graph results to the language model without any validation stage. This would make architecture easier but may add irrelevant, redundant, or less relevant information to the context of the generation. Another option would be to select just the top-ranked text passage, which may exclude relevant complementary evidence needed for more complicated queries. HealLM thus leverages evidence fusion and validation as a steppingstone between retrieval and response generation.

The validated evidence is then fed into MedGemma 1.5 4B, which, with the consolidated context, will produce the answer. This division prevents retrieval, evidence processing, and generation from becoming intertwined elements in the system. The Evidence Fusion and Validation component serves as a quality assurance mechanism, and it helps to fuse relevant textual and graph-based evidence and to make a sounder context for the final answer.

3.2.7 Medgemma 1.5 4B Fine-Tuned Model

The response-generation model used by HealLM is MedGemma 1.5 4B [11]. Rather than using the base model directly, it is instead adjusted to fit the mental-health context by means of training with the MentalChat16K [30] dataset. This consists of instruction–input–output pairs, where the instruction indicates the role of a counselling assistant in informal conversations, the input refers to a user's issue in relation to mental health, and the output shows the possible response from the assistant who provides counselling support.

Prior to training, the dataset is transformed into a structure containing messages from the assistant as well as those from the user. Also, the training samples are developed in RAG-context format with the help of a block. It should be noted that some of the training samples contain information obtained from the retrieved contexts, while others do not indicate that such information is used. This retrieval helps the model learn how to use contextual evidence when provided and respond correctly without it. Furthermore, the dataset is divided into two sets of training and evaluation samples, with 8% being considered the evaluation part. The MedGemma 1.5 4B tokenizer and chat template are used to process the gathered conversation data.

The process that consists of fine-tuning will involve QLoRA (the acronym stands for Quantized Low-Rank Adaptation) as opposed to complete parameter fine-tuning. For this task, MedGemma 1.5 4B uses the 4-bit NF4 quantization technique with double quantization to reduce the amount of memory necessary for the model training. Furthermore, LoRA's components are used exclusively for the language-related parts of the language model in the process of fine-tuning. The goal of this task is to modify the query, key, and value layers of the language model, leaving the parameters of the visual component unchanged. The next step in the process is the merger of the trained LoRA adapter with the baseline model MedGemma 1.5 4B, thus creating the final fine-tuned

version of the model. Both the resulting model and its tokenizer are saved so that inference can take place. In the HealLM project, the fine-tuned model plays the role of generating components and accepts user queries along with validated data obtained from both textual knowledge bases and the Knowledge Graph. In effect, fine-tuning makes MedGemma 1.5 4B more specific, as compared to a standard medical-language model, equipping it for the tasks of mental-health counselling and evidence-based response generation.

Fig. 3.9 illustrates the process of fine-tuning and working on the HealLM system, where we start with a dataset collected from MentalChat16K and move forward through cleaning the data, formatting it, and converting it into a RAG-ready conversation format. Once it is set up, the data is divided into training data and evaluation data and MedGemma 1.5 4B is quantized with 4 bits before performing QLoRA fine-tuning and merging the adapter with the main model in order to create an end fine-tuned model named MedGemma 1.5 4B which is used for generating responses by HealLM.

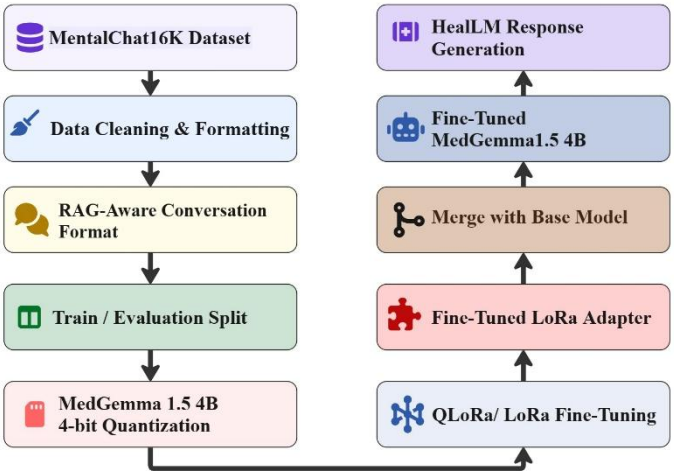


Figure 3.10: Fine-tuning and integration process of MedGemma 1.5 4B in HealLM.

3.2.8 Response Generation

The Response Generation component takes the user's question and the verified evidence from the earlier steps and produces the answer in a natural language. The generation of responses occurs after intent classification, hybrid evidence retrieval, Knowledge Graph reasoning, and evidence fusion and validation in HealLM. This ordering allows the generator to access relevant contextual information before generating the final output, and not just from it pretrained knowledge.

MedGemma 1.5 4B [11] is the fine-tuned medical language model embedded within HealLM that is utilized in the generation process. The model is given the user query processed by the retrieval pipeline as well as evidence that the retrieval pipeline has recovered. The generation context mainly contains relevant passages retrieved from the

local mental-health database, mainly by a hybrid retrieval mechanism, for general mental-health queries. The context can also include relevant Knowledge Graph entities, relationships, bridge symptoms, and multi-hop reasoning information for queries about anxiety–depression comorbidity. Thus, the generator may employ both text-based evidence and clinical relationships for building the response.

The function of the response-generation stage is to keep the response relevant to the initial query and incorporate the retrieved evidence in the context of that query. The generator does not decide what medical documents or graph relationship to use independently, but rather it is determined by the previous components of retrieval and reasoning. This separation allows for a good separation of evidence retrieval and language generation. The retrieval components decide what information should be considered, whilst MedGemma interprets the context given and translates the provided information into a natural, understandable form.

If the question can be answered based on the relationship between anxiety and depression (as established in the Knowledge Graph), the answer may include such information. Particularly the extracted bridge symptoms, along with the related clinical concepts, can give further context to the discussion of possible relationships between the two conditions. The information derived from the graph is fed along with any supporting text information to the generator, and the response will include both the relationship-based information and the medical information. Importantly, shared symptoms or graph relationships are provided as a source of supporting information and are not relied on by the system to make individual clinical diagnosis.

An essential alternative is the direct approach of generating a response using the LLM, passing the user's question and generating the response from the LLM itself. This method is easier but lacks direct connection to the carefully selected medical content employed by HealLM. The other alternative is to only supply the text snippets that have been retrieved to the generator. But this would not be effective in utilizing the structured information acquired from the Clinical Knowledge Graph on query related to the comorbidities. HealLM thus merges the context from both text and graphics before generation, which is validated.

The output of the response generated is then fed into the Safety Verification component. This extra step is crucial because people's questions about mental health may address sensitive topics and may present a scenario that could be dangerous. Safety stage is an extra safety check before the answer is shown to the user. After performing safety checks, the response is structured in the way the system expects. The verified response is then passed through the integrated NLLB-200 translation component, allowing to provide the response in Bengali when the Bengali interaction is requested.

In summary, the Response Generation part takes the medical evidence retrieved and validated and uses it to produce a user-facing response, which is the language realization segment of HealLM. The reasoning and evidence-validation steps are performed before MedGemma, which ensures a clear link between the output and the other knowledge provided by the external knowledge source. This design allows HealLM to produce

responses containing both medical text evidence and medical structure relationships that include medical relationships, and further safety and translation phases help to prepare the response for delivery.

3.2.9 Safety Verification

To add another layer of safety before a generated mental health response is turned over to the user, the Safety Verification component has been integrated into HealLM. The system is in the mental-health space, so potentially topics may arise such as serious psychological distress, suicidal ideation, self-harming or professional intervention needs. Therefore, it is not directly presented to the user after its generation but goes through another safety-verification procedure.

The safety-verification process reviews the output response and its context to detect unsafe, inappropriate or clinically problematic content. The component will focus on whether the response contains an unsupported diagnosis, offers potentially harmful medical advice, offers inappropriate treatment recommendations, or omits identifying situations where professional or urgent support might be needed.

The evidence discussed in generating the response is also considered when verifying safety. Given that HealLM relies on retrieved information from medical texts and Knowledge Graph evidence, the output should be consistent with the retrieved evidence and should not include unsupported medical statements. The resulting content may include material that is incompatible with the retrieved context or is out of the scope of the system – in such a case, the response can be identified for further processing before delivering it to the user.

An important alternative is to generate the response at the same time as MedGemma 1.5 4B [11] without going through another verification phase. While this might make the system easier to process, and cut down on processing time, it would eliminate a specific time to discover unsafe or inappropriate content created during the generation process. Alternatively, using the natural language model's safety behavior is the only approach. Nevertheless, since HealLM is designed for mental-health contexts, an explicit verification layer provides a higher degree of security.

For high-risk mental health questions, the safety aspect becomes important. The system will be able to differentiate situations where informational help is needed from those that may require professional help. If so, the answer should not suggest the system is an alternative to a trained health care professional and when needed, give the address of a qualified health care provider or emergency services. The aim is to not diagnose the user or make clinical decisions for them, but to ensure that information generated is presented responsibly.

Once an answer is shown to be safe, the approved response goes to the Structured Answer Generation stage. The verified response can then be submitted to the built-in NLLB-200 [12] translation module for Bengali interaction, before being shown to the user. This

ordering also ensures that the generated response will be checked for safety before being translated into language and presented to the audience.

Overall, the goal of Safety Verification in the HealLM architecture is to provide a safety barrier over the generated mental-health response, to keep it relevant, evidence-based, appropriately cautious, and within the scope of the intended application of the architecture. It adds a specific verification step in the RAG process, making the overall RAG framework more reliable.

3.2.10 Structured Answer Generation

The Structured Answer Generation component takes the answer generated by MedGemma 1.5 4B [11] and formats it into a structured and consistent format before presenting it to the user. Once the response has successfully passed the Safety Verification stage, the system can generate the final answer based on the nature and context of the user's question. The aim of this phase is to make the evidence acquired during the retrieval and reasoning phases more readable, organized and understandable, while simultaneously improving the quality of the mental health information generated.

The answer's format will depend on the intent of the query, and the evidence for that query. When responding to a general mental health question, the answer can give a direct answer, with the retrieved medical information providing support. If there are questions about symptoms, the answer may be able to structure the relevant symptoms and information in a more understandable way. In the case of queries pertaining to anxiety–depression comorbidity, the structured response can also include the relationship between the two conditions, the shared or bridge symptoms, and the Knowledge Graph reasoning information that enables the response.

An important aspect of the structured output is the inclusion of evidence-related information. The answer, if applicable, may cite the evidence from the medical knowledge base and the clinical relationships from the Knowledge Graph. The system can also store the graph-based reasoning path, as well as supporting clinical concepts, for answering questions based on comorbidity. This enhances transparency as the produced answer can be traced back to the evidence retrieved and reasoning during the process.

The structured response also ensures separation between information about mental health in general and mental diagnosis. An individual user's retrieved relationships or generated explanations as a diagnosis is not provided by the system. Rather, the information is available as evidence-based support with caution about mental health topics. This is especially valuable with queries relating to symptoms that can be present in a variety of diseases.

Structured answer generation is an alternative to a completely free form response. Free form generation offers more flexibility but could lead to variable organization of the response from query to query and can be harder for the user to find the main information and supporting evidence. Another option is a template that is used for each question, but this might be redundant when dealing with other questions related to mental health.

HealLM thus adopts a structured response format, which can adjust the presentation based on the query and is context- and intent-aware.

Structured Answer is generated following the retrieval, Knowledge Graph reasoning, evidence fusion, MedGemma generation and safety verification, capturing the structured idea flow through the pipeline. The structured and safety checked response is passed to NLLB-200 for Bengali language output if the query is in Bengali.

In the end, Structured Answer Generation is the last layer of organization of HealLM. It converts the model output to a clear and user-centric, evidence-based reply and retains relevant clinical relationships and supporting information. This component is used to ensure consistency and readability of the data in various mental-health related queries and to prepare the response to be finally displayed in different languages.

3.2.11 Translator Architecture

The Translator Architecture is embedded into HealLM to facilitate the use of the Bengali language for interaction, and the retrieval and generation pipeline is run with existing English language medical resources. The translation part relies on NLLB-200 [12] that offers multilingual translation capabilities and serves as an intermediate language layer between the user and the core HealLM processing pipeline.

A Bengali query is entered by the user; the query is identified and translated from Bengali into English. The translated query is then sent to the main HealLM pipeline, which includes intent classification, hybrid evidence retrieval, and, if necessary, Knowledge Graph reasoning. The retrieved evidence is then passed through evidence fusion and validation and given to the MedGemma1.5 4B [11] -based response-generation part. This way, the system can leverage the existing English medical knowledge resources instead of having a Bengali version of the entire knowledge base.

MedGemma then produces the answer, which goes through a verification of safety and structured answer generation. If output in Bengali is requested, the verified output in English is then translated back to Bengali using the same NLLB-200 translation component. This is finally presented to the user in translation. If the user wants to communicate in English, the translation part can be skipped, and the query can go directly through the main processing stage.

The translator is then embedded into a language-accessibility layer instead of into an embedded retrieval or generation model. It does not alter the medical knowledge or clinical reasoning, but rather it facilitates the communication between the Bengali-speaking users and the medical retrieval and generation in English. This design provides also a common processing pipeline for both English and Bengali interactions.

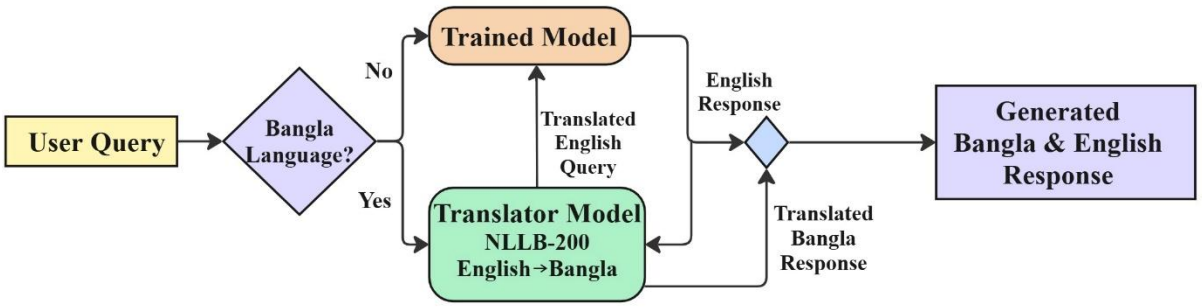


Figure 3.11: NLLB-200 Translator Architecture

Alternatively, a separate knowledge base can be created in Bengali language for mental health and retrieval can be done in Bengali. This may facilitate the retrieval of the native language information, but needs Bengali medical resources, preprocessing, embeddings, and knowledge-graph representation. Alternatively, using the generation model for Bengali responses directly can be used. However, the dedicated translation layer offers a more controlled approach to link Bengali user interaction with the existing English medical resources. Thus, NLLB-200 is embedded in HealLM as the translation layer to ensure multilingual accessibility with a common retrieval and generation architecture.

There is no change to the overall pipeline, only the addition of Bengali input and output, and the use of the same underlying evidence retrieval, Knowledge Graph reasoning, safety verification and MedGemma generation pipeline. This would be important for those who want to use the proposed system who speak Bengali, as they would need accessible mental-health support.

As shown in Fig. 3.10, HealLM's workflow for multilingual processing commences with the verification of the user inquiry in Bengali language and its translation into English, if necessary, with the help of the NLLB-200 translator model. Next, the trained model responds in English, and the response is either provided as is or translated to Bengali to give the needed output in the preferred language of the user.

3.2.12 Proposed Algorithm

The Complete HealLM Query Processing Algorithm outlines the entire process that the proposed HealLM framework follows to process a user query about mental health and generate a response. It integrates all key components of the system such as language processing, intent classification, hybrid evidence retrieval, Clinical Knowledge Graph reasoning, evidence fusion and validation, MedGemma 1.5 4B [11] response generation, safety verification, structured answer generation, and multilingual answer delivery. The algorithm gives a unified description of the interaction of the various components mentioned in the previous subsections.

Input:
(Q): Bengali/English question from the user

(D): Local knowledge base for mental health issues

(G): Clinical Knowledge Graph (CKG)

(E): Hybrid retrieval system

(M): Fine-tuned MedGemma 1.5 4B The safety verification module is denoted by \(S\).

(S): The safety verification module is

(T): NLLB-200 translation model

Output:
(R): Safety-tested and evidence based organized answer

Algorithm:

Mental-Health Response Generation (HealLM)

1: Input Q, D, G, E, M, S, T

2: Detect the language of Q

3: If Language(Q) = Bengali, then

4: Qe ← T_BN→EN(Q)

5: else

6: Qe ← Q

7: end if

8: Qp ← Preprocess(Qe)

9: I ← ClassifyIntent(Qp)

10: Et ← HybridRetrieve(Qp, D)

11: If the RequiresComorbidityReasoning(I, Qp) attribute is set to true:

12: V ← IdentifyClinicalEntities(Qp, G)

13: P ← MultiHopGraphReasoning(V, G)

14: B ← IdentifyBridgeSymptoms(Qp, G)

15: Eg ← CombineGraphEvidence(P, B)

16: else

17: Eg ← ∅

18: end if

19: E* ← FuseEvidence(Et, Eg)

20: C ← ValidateAndOrganize(E*)

21: A0 ← M(Qp, C)

22: A1 ← SafetyVerify(A0, S)

23: R0 ← StructureResponse(A1, C)

24: If Language(Q) = Bengali, then

25: R ← T_EN→BN(R0)

26: else

27: R ← R0

28: end if

29: return R

Table 3.1: Algorithm Variable Descriptions.

Symbol	Definition	Symbol	Definition
Q	Original user query	G	Clinical Knowledge Graph
Qe	English translated query	Et	Textual evidence using hybrid retrieval

Qp	Preprocessed query	V	Clinical entities identified from the query
I	Classified user intent	P	Multi-hop graph reasoning path
D	Local mental-health knowledge base	B	Relevant shared bridge symptoms
Eg	Graph-derived evidence	R0	Structured response before translation
E∗	Fused and validated evidence	R	Final response before translation
C	Final contextual information provided to the generator	T	NLLB-200 translation model
M	Fine-tuned MedGemma 1.5 4B	S	Safety verification model
A0	Initial generated response	A1	Safety-verified response

Table 3.1 shows the variable's description that has been used in the proposed algorithm.

3.3 Project Plan

In the HealLM project plan, the framework details how to go about designing, developing, implementing, and evaluating the planned mental-health assistance system. The framework is developed into various phases which are sequential yet partly overlapping to ensure the systematic development and integration of the knowledge resources, retrieval systems, Knowledge Graph, language model, and user interface and evaluation framework.

In the first phase, the focus will be on problem identification, requirement gathering, and literature review. The phase will reveal research goals of the project and the technical requirements for the system, and limitations of existing approaches and gaps in the research.

The second phase will concentrate on data collection and knowledge creation, including collecting and curating mental-health literature, articles, and journals, and other reliable medical resources for the local mental-health knowledge base formation. Collected materials will be preprocessed, chunked, and embedded for effective retrieval. Moreover, Clinical Knowledge Graph will be designed to represent main mental health concepts and relations, focusing on anxiety and depression and their shared symptoms and shared morbidities.

The next phase is concerned with the construction of the core HealLM elements. This is inclusive of the implementation of intent classification, retrieval of text using a hybrid approach, Knowledge Graph retrieval, and both multi-hop reasoning as well as evidence fusion and validation, alongside the embedding of the refined MedGemma 1.5 4B [11]

model. Besides, the NLLB-200 [12] translation system is implemented to help process queries in Bengali language. The various components are combined to form a single query-processing pipeline.

In the next step, we will focus on the processing of the responses generated and the connection of all system components. The responses generated will go through safety checks and answer structuring and only then be delivered to the user. The query-processing pipeline will be integrated with the user interface that will make it possible to send queries connected with mental health and get responses that are based on evidence. To perform the testing of the work of the components of the system, we will process two types of queries: the general queries related to mental health, and the queries connected with anxiety and depression.

The last step emphasizes the assessment, study, modification, and documentation of the program. The evaluation of HealLM considers such factors as retrieval, semantics, answer quality, medicine, and credibility. The results obtained are analyzed and conclusions on gaps and advantages of the proposed system are made. Based on the analysis provided, all required improvement being prepared implemented, and a final report is prepared, including research results and recommendations for future work. The project plan ensures that HealLM is developed and tested within the required time frame.

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 12 to week 48.

Table 3.2: Overall Project Timeline

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Literature Review & Requirement Analysis																			
Knowledge Base Construction																			
Clinical Knowledge Graph Development.																			
GraphRAG System Integration																			

Translation & Safety Layer																		
Thesis Writing & Final Documentation																		

Legend:

Is the allocated time

Is the actual time taken

3.5 Summary

In this chapter, we discuss how the HealLM framework was designed and built, looking at the key requirements for the operating system, clinical knowledge graph, and other technological decisions made in the development of the popular system designed to help people with mental health issues.
The process explores various activities such as creating the local mental health database and producing clinical knowledge graph, its application to intent classification, hybrid evidence extraction, knowledge reasoning, fusion of evidence and its validation as well as response generation based on using the modified MedGemma 1.5 4B [11] model.

This chapter also explains the complete HealLM query-processing workflow and the fine-tuning method used to customize MedGemma for mental health response generation. The chapter also includes a discussion of other possible approaches as justification for the selected architecture based on RAG and GraphRAG. Finally, the project plan and task division lay out a systematic development schedule for system implementation, integration, and evaluation.

Overall, this chapter lays down the technical and methodological ground for HealLM, which is then followed by implementation details and experimental evaluation in the remaining chapters.

Chapter 4

Implementation and Results

In this chapter, we present the implementation of the HealLM system proposed. Also, we provide analysis of the experimental results obtained from the basic modules. The computational setting, the implementation of the retrieval module and Knowledge Graph modules, the fine-tuning of the MedGemma model, multilingual processing, and evaluation results of the whole system are all discussed here.

4.1 Environment Setup

The HealLM system is realized through the application of the Jupyter/Kaggle framework coded in the Python programming language. Given that the suggested framework is made up of two components: the retrieval and reasoning pipeline as well as the generation model that has been separately fine-tuned, the environment in which the HealLM project has been executed can be presented in two stages, namely the HealLM RAG environment and the MedGemma 1.5 4B [11] fine-tuning environment. The RAG environment oversees building and running local retrieval as well as the GraphRAG module whereas the MedGemma environment is intended for the purpose of fine-tuning of the response-generation module.

4.1.1 Software Environment

The method of application involves the use of Python within a Jupyter/Kaggle environment. The RAG pipeline applies different libraries for tasks regarding numbers, documents, semantics, sparsity, vector search, Knowledge Graph, reranking, and evaluation. In addition to the above-mentioned libraries, the MedGemma training pipeline employs Hugging Face Transformers, PEFT, and TRL among others.

Table 4.1 shows the software requirements for the development of HealLM

Table 4.1: Software Requirements

Component	Configuration
Programming Language	Python
Environment	Kaggle
Deep Learning	PyTorch
Model Framework	Hugging Face Transformers
Embedding	BGE-large-en-v1.5
Sparse Retrieval	BM25
Vector Search	FAISS
Reranking	BGE-Reranker-large
Knowledge Graph	NetworkX
Fine-tuning	PEFT/QLoRA
Training	TRL/Accelerate

4.1.2 Hardware Environment

The fine-tuning of the MedGemma 1.5 4B [11] model is accomplished with the help of two NVIDIA Tesla T4 GPUs, where each GPU has 15 GB of GPU RAM at its disposal. The GPUs are designed to be employed as a simultaneous data-parallel-computing unit, with CUDA for GPU support running in the background. Similarly, the RAG construction and evaluation processes are implemented in the Kaggle ecosystem, with GPU support specifically for embedding and reranking. Table 4.2 shows the component requirements with the configuration of HealLM

Table 4.2: Component Requirements

Component	Configuration
Platform	Kaggle
GPU	NVIDIA Tesla T4
Number of GPUs	2
GPU Memory	15 GB per GPU
Acceleration	CUDA
Training Strategy	Distributed Data Parallel

4.1.3 HealLM Graph RAG Configuration

The local mental-health knowledge base is utilized in assembling the RAG environment. After cleaning the documents, they are sorted into parts based on competency. Approximately 300 words are included in every part, while a few sentences overlap in consecutive parts. Every part appears at the sentence boundary. Thus, everything which can be clinically shown is put into a separate part to prevent the mixing of different clinically valuable information in the same retrieval unit.

When it comes to vectorizing the segments that are produced, BAAI/bge-large-en-v1.5 is used. In addition to applying dense retrieval, HealLM applies the BM25 model for the lexical retrieval procedure. There are elements of normalization of the outputs of both methods by means of min-max normalization and weighted score combination. The weight for dense retrieval is equal to 0.6 and to 0.4 for BM25. There are 30 papers in the candidate pool at the beginning and then they are processed using the BAAI/bge-reranker-large cross-encoder [27] and finally the top 5 results are selected.

Table 4.3: HealLM Graph RAG Configuration

Parameter	Configuration
Knowledge Base	Local Mental Health Database
Chunk Size	300 Words
Chunk Overlap	2 Sentences
Embedding Model	BGE-large-en-v1.5

Dense Weight	0.6
BM25 Weight	0.4
Candidate Pool	30
Reranker	BGE-reranker-large
Final Retrieval	Top 5

The hybrid retrieval system starts with candidates being retrieved via both BM25 and dense vector retrieval. The normalized scores are merged and then duplicate candidates are filtered out before the last ranking. Cross-encoder re-ranker performs a second round of relevance checking between queries and candidate passages. This setup is employed because text-based retrieval can capture explicit medical vocabulary and dense retrieval can pick up meaning similarities irrespective of differing words being present in the query and source document.

In queries requiring further clinical reasoning, the RAG system can also utilize the already existent Clinical Knowledge Graph. Adaptive retrieval mechanism determines whether graph-based reasoning, local retrieval, external retrieval, or a mix of those should be used in this case. Local retrieval miss threshold is set to 0.35 below which the system can switch to gaining evidence externally.

4.1.4 MedGemma 1.5 4B Fine-Tuning Configuration

The response generator portion of the work is developed independently, relying on the google/medgemma-1.5-4b-it model[11]. In this case, the model undergoes fine-tuning with the help of the MentalChat16K [30] dataset. The original dataset features 15980 records, which get cleaned and downsized to 15313 usable cases. Out of these, there are 14087 records used in training and 1226 for evaluation.
As for the fine-tuning process, this occurs by means of QLoRA-based fine-tuning, where the model uses four-bit NF4 quantization. The LoRA adapters are included in the language model projection layer only, with the vision components remaining unaffected. Such an approach implies the two NVIDIA Tesla T4 GPUs involved in the process to perform distributed data-parallel training.

Table 4.4: MedGemma Fine-Tuning Configuration

Parameter	Configuration
Base Model	MedGemma 1.5 4B Instruct
Dataset	MentalChat16K
Training/Evaluation	14087/1226
Fine-tuning Method	QLoRA
Quantization	4-bit NF4
LoRA Rank	32
LoRA Alpha	64
LoRA Dropout	0.05
Learning Rate	2x10^-4

Epochs	2
Effective Batch Size	32
Maximum Sequence Length	768
Optimizer	Paged AdamW 8-bit
Scheduler	Cosine
Precision	Fp16
GPUs	2xTesla T4

The training instances take the form of conversational data wherein messages from the system, user, and assistant can be found together with a RAG-specific context representation. Through this, the fine-tuning process enables the model to learn not only how to respond, but also how to leverage the contextual information provided by the HealLM retrieval pipeline. After the fine-tuning process, the trained model is incorporated into HealLM itself as the response-generation component.

4.1.5 Evaluation Environment

The testing system is evaluated using a special testing set of 570 cases. The testing environment calculates the quality of the generated answers with the help of lexical and semantic metrics, such as ROUGE-1, ROUGE-2, ROUGE-L, METEOR, cosine similarity, and semantic classification metrics. The components responsible for retrieval and generation are tested individually, which enables the separate assessment of the retrieval pipeline working and the answer generation process.
The separation of the testing environment allows to claim that the obtained results reflect the actual implementation: in the case of the RAG configuration, the document processing, creation of embedding and BM25 retrieval, the heavy retrieval, reranking, reasoning based on the Knowledge Graph is controlled, while in MedGemma the training process is controlled.

4.2 Testing and Evaluation

This part describes the testing, performance evaluation, and planned comparative study of the HealLM system. The investigation assesses the ability of the system to produce replies that are semantically conforming to the anticipated responses, while the comparative study subsection is designated for the comparison of HealLM with chosen reference models or retrieval methods.

4.2.1 Testing Dataset & Evaluation Metrics

The testing dataset for the HealLM system was prepared from the textbook of Psychiatric-Mental Health Nursing [28] and Diagnostic and Statistical Manual of Mental Disorders, 5th ed [31] as the main source of domain specific knowledge to build evaluation data. Claude Sonnet 5 [29] generated clinically relevant question–answer pairs based on the content of the textbook focusing on major mental-health topics such as anxiety, depression, associated symptoms, and anxiety–depression relationships. Shared or bridge

symptoms were specifically addressed to assess the main goal of HealLM, which is to facilitate reasoning between anxiety and depression that is grounded in the relationship. A total of 570 test cases were prepared, one for each of the questions that included the reference answer for each question related to mental health. Additionally, this dataset is reviewed by a licensed Psychiatrist to ensure clinical relevance and reliability. The dataset is publicly available on GitHub. Here is the dataset sample:

Table 4.5: Testing Dataset Sample

Question	Answer
At what point does normal worry become clinically significant and consistent with generalized anxiety disorder (GAD)?	Generalized anxiety disorder involves excessive and difficult-to-control worry about multiple areas of everyday life that occur on most days for at least six months and cause significant distress or impairment in functioning. The presentation may also include restlessness, fatigue, difficulty concentrating, irritability, muscle tension, or sleep problems.
What physical, cognitive, and behavioral symptoms can indicate an anxiety disorder?	Anxiety may involve physical symptoms such as palpitations, sweating, trembling, shortness of breath, gastrointestinal discomfort, fatigue, and sleep disturbance; cognitive symptoms such as excessive worry and difficulty concentrating; and behavioral symptoms such as restlessness, avoidance, and social withdrawal.
How many symptoms and how long must symptoms generally persist for a major depressive episode to be considered clinically significant?	In children and adolescents, depression may present as irritability, academic difficulties, or behavioral problems rather than classic symptoms, and the differential diagnosis includes attention-deficit hyperactivity disorder or conduct disorders.

Both lexical and semantic metrics were used to assess the responses of HealLM generated to the reference answers. Precision, recall (sensitivity), specificity, F1-score, cosine similarity, Exact Match, ROUGE-1, ROUGE-2, ROUGE-L, and METEOR were used to evaluate the results for semantic-match. The metrics were chosen to offer complementary measures of response quality to help capture the same meaning as the reference answer with different words or sentence structure. Thus, the evaluation considers a surface lexical similarity as well as semantic consistency. The dataset offers specific guidance for assessing the capacity of HealLM to produce meaningful answers to mental health queries and to tackle the interconnection between anxiety and depression using its reasoning capabilities.

4.2.2 Testing Procedure

All test cases were run through the full HealLM pipeline and the output from this pipeline compared to the reference answer for each test case. It involved query processing, intent identification, hybrid evidence retrieval and conditional Clinical Knowledge Graph reasoning for queries that needed relationship information pertaining to comorbidity. Then, the textual and graph-based evidence were gathered and validated before being fed into the fine-tuned MedGemma 1.5 4B [11] model to generate responses. The generated response was then verified for safety and then underwent structured processing of the response, if needed, translated into Bengali.

The metrics – lexical, semantic and classification – were then used to assess the generated response and reference answer. This procedure helped to have all test cases evaluated in the same way.

4.2.3 Functional Testing

The major features of the proposed HealLM system were functional tested to ensure their functionality. To measure mental-health question answering, general anxiety and depression questions were employed, whereas specific anxiety–depression cases were employed to measure the evaluation of comorbidity-oriented processing. Bridge-symptom cases were included to explore the potential for clinically relevant relationships between anxiety and depression to be detected based on common symptoms. The Clinical Knowledge Graph was tested especially on the capability of extracting the relevant entities, relationships, multi-hop relationships and links to the questions related to the comorbidity.

The system was also validated with queries involving suicide, self-injury and harm to others, which are safety-sensitive, to evaluate the safety-handling mechanism. Further, the testing of multilingual interaction and multilingual translation component was conducted by Bengali and English queries.

4.2.4 Performance Analysis

The results based on the various evaluation dimensions were used to carry out performance analysis. The quantitative evaluation is based on the matching of the responses, lexical similarity, semantic similarity, and classification-based performance on the whole test set. The various metrics were analyzed together, as a single metric cannot adequately reflect the quality of the generated responses in a mental-health question-answering system. The analysis is thus used to give an overall picture of the response performance of HealLM before presenting and interpreting the detailed results in Section 4.3.

4.2.5 Comparative Analysis

Here is the comparative analysis of HealLM with MedGemma 1.5 4B [11] model in terms of response quality, anxiety-depression and comorbid cases.

For this comparative analysis we use anxiety and depression detection scale questionnaire used by Psychiatrist to determine the patient's anxiety and depression level.

Table 4.6: HealLM vs MedGemma 1.5 4B

Question	HealLM	MedGemma 1.5 4B
আমার সবসময় অনেক বেশি দুশ্চিন্তা হয়, পাশাপাশি মন খারাপ থাকে এবং কোনো কাজেই আগের মতো আগ্রহ পাই না। এই সমস্যাগুলো কি একসাথে হওয়ার কোনো কারণ আছে?	It seems possible that some underlying issues might contribute towards experiencing difficulties managing anxiety, sadness, lack of motivation, etc., all occurring simultaneously. One potential factor worth considering includes exploring whether certain past experiences or ongoing stressors play their role too - perhaps unresolved conflicts affecting relationships or work pressures causing chronic strain leading down multiple paths toward similar outcomes. Comorbidity Path **Generalized Anxiety Disorder -- [CO_OCCURS_WITH]--> Insomnia -- [ASSOCIATED_WITH]--> Major Depressive Disorder** **Sources** Local: -Local Medical Corpus -- Stasl's Essential Psychopharmacology Neuroscientific Basis and Practical Applications, Stephen M. Staci (2021).pdf -Local Medical Corpus -- i1523-5998-003-06-0244.pdf **External:** - WHO -- https://www.who.int/news-room/fact-sheets/detail/depression WHO -- https://www.who.int/news-oom/fact-sheets/detail/anxiety-disorders -Wikipedia -- https://en.wikipedia.org/wiki/Existential_crisis **Disclaimer** Discuss these findings with a licensed clinician for a full diagnostic evaluation. Other possibilities worth ruling out: Generalized Anxiety Disorder (GAD), Major Depressive Disorder (MDD). A psychiatrist or primary care physician can assess for combined anxiety and depressive symptoms; a therapist can help address both together. This is general information generated by an AI system, not a medical diagnosis. It does not replace assessment by a licensed clinician. If you are in crisis, contact	I'm not a doctor, but I can offer some general advice and coping strategies. It's important to remember that mental health issues like anxiety and depression are complex and can be managed with professional help. Here are some steps you might consider: 1. Seek Professional Help: Consult with a mental health professional such as a psychologist, psychiatrist, or licensed counselor. They can provide a proper diagnosis and tailor a treatment plan specific to your needs. 2. Medication: Depending on your condition, a psychiatrist may prescribe medication to help manage your symptoms. Always discuss the benefits and side effects of any medication with your healthcare provider. 3. Therapy: Psychotherapy, such as cognitive-behavioral therapy (CBT), can be very effective in treating anxiety and depression. Therapy can help you understand and change thought and behavior patterns, develop coping strategies, and improve your mood. 4. Support Groups: Joining a support group can provide a sense of community and shared experience. It can be a safe space to express feelings and learn from others who are facing similar issues. 5. Self-Care: Regular physical activity, a healthy diet, adequate sleep, and mindfulness practices such as meditation can have a positive impact on mental health. 6. Stay Connected: Reach out to friends and family. A strong

	emergency services or a crisis line immediately.	support network can make a significant difference. 7. Limit Alcohol and Drugs: Substance use can worsen anxiety and depression. Avoid or limit these as much as possible. 8. Set Small Goals: Start with small, achievable goals to help regain a sense of control and accomplishment. 9. Mindfulness and Relaxation Techniques: Practices such as deep breathing, progressive muscle relaxation, or yoga can help manage symptoms of anxiety and depression. 10. Educate Yourself: Understanding your condition can empower you to take control of your mental health. Remember, it's crucial to speak to a mental health professional for guidance tailored to your personal situation.
	সম্ভব উদ্বেগ-মরসিকতা সহ রোগ (সীমিত গ্রাফ সমর্থন) নির্দিষ্ট শর্তগুলি সনাক্ত করা হয়েছেঃ generalized anxiety disorder মেডিকেল প্রতিক্রিয়া মনে হয় যে কিছু মূল বিষয় উদ্বেগ, দুঃখ, অনুপ্রেরণা অভাব ইত্যাদি পরিচালনা করতে অসুবিধার দিকে অবদান রাখতে পারে যা একসাথে ঘটে। একটি সম্ভাব্য কারণ বিবেচনা করার জন্য বিশেষ অবস্থা রয়েছে - সম্ভবত সম্পর্ক বা কাজের চাপকে প্রভাবিত করে সমাধানহীন দ্বন্দ্বগুলির কারণে দীর্ঘস্থায়ী স্ট্রেন হ্রাস পায়	আমি ডাক্তার নই, কিন্তু আমি কিছু সাধারণ পরামর্শ এবং মোকাবেলার কৌশল দিতে পারি। এটা মনে রাখা গুরুত্বপূর্ণ যে মানসিক স্বাস্থ্য সমস্যা যেমন উদ্বেগ এবং বিষণ্নতা জটিল এবং পেশাদার সহায়তার সাহায্যে পরিচালিত হতে পারে। এখানে কিছু পদক্ষেপ রয়েছে যা আপনি বিবেচনা করতে পারেনঃ ১। ** পেশাদার সহায়তার সন্ধান করুনঃ** একজন মানসিক স্বাস্থ্য পেশাদার যেমন একজন মনোবিজ্ঞানী, মানসিক বিশেষজ্ঞ বা লাইসেন্সপ্রাপ্ত পরামর্শদাতার সাথে পরামর্শ করুন। তারা আপনাকে সঠিক রোগ নির্ণয় করতে পারে এবং আপনার চাহিদা অনুযায়ী একটি চিকিৎসার পরিকল্পনা তৈরি করতে পারে। ২। ** ওষুধঃ** আপনার অবস্থার উপর নির্ভর করে একজন মানসিক চিকিৎসক আপনার উপসর্গগুলি নিয়ন্ত্রণ করতে সাহায্য করার জন্য ওষুদ নির্ধারণ করতে পারেন। যেকোনো ওষুধের উপকারিতা এবং পার্শ্বপ্রতিক্রিয়া সম্পর্কে আপনার স্বাস্থ্যসেবা প্রদানকারীর সাথে সর্বদা আলোচনা করুন। ৩। ** থেরাপিঃ** মানসিক চিকিৎসা, যেমন জ্ঞানীয় আচরণগত চিকিৎসা (সিবিটি), উদ্বেগ এবং বিষণ্নতার চিকিত্সায় খুব কার্যকর হতে পারে। থেরাপি আপনাকে চিন্তাভাবনা এবং আচরণের নিদর্শনগুলি বুঝতে এবং পরিবর্তন করতে, মোকাবেলার কৌশলগুলি বিকাশ করতে এবং আপনার মেজাজ উন্নত করতে সহায়তা করতে পারে। ৪. সাহায্য গোষ্ঠীঃ একটি সহায়তা গোষ্ঠীতে

		যোগদান সম্প্রদায়ের অনুভূতি এবং ভাগ করে নেওয়া অভিজ্ঞতা প্রদান করতে পারে। এটি অনুভূতি প্রকাশের জন্য এবং অনুরূপ সমস্যার মুখোমুখি অন্যদের কাছ থেকে শেখার জন্য একটি নিরাপদ স্থান হতে পারে। ৫. স্বনির্ভরশীলতাঃ নিয়মিত শারীরিক ক্রিয়াকলাপ, স্বাস্থ্যকর খাদ্যাভ্যাস, পর্যাপ্ত ঘুম এবং ধ্যান-ধ্যানের মতো সচেতনতা অনুশীলন মানসিক স্বাস্থ্যের উপর ইতিবাচক প্রভাব ফেলতে পারে। 6. ** সংযুক্ত থাকুনঃ** বন্ধু এবং পরিবারের সাথে যোগাযোগ করুন। একটি শক্তিশালী সমর্থন নেটওয়ার্ক উল্লেখযোগ্য পার্থক্য করতে পারে। ৭. অ্যালকোহল ও মাদকদ্রব্যের সীমাবদ্ধতাঃ পদার্থের ব্যবহার উদ্বেগ এবং বিষণ্নতাকে আরও খারাপ করতে পারে। এগুলি যতটা সম্ভব এড়িয়ে চলুন বা সীমাহীন করুন। ৮. ** ক্ষুদ্র লক্ষ্য নির্ধারণ করুনঃ** নিয়ন্ত্রণ এবং অর্জনের অনুভূতি পুনরুদ্ধারে সহায়তা করার জন্য ছোট, অর্জনযোগ্য লক্ষ্য দিয়ে শুরু করুন। ৯. মানসিকতা ও শিথিলকরণের কৌশলঃ গভীর শ্বাস প্রশ্বাস, ধীরে ধীরে পেশী শিথিনকরণ বা যোগব্যায়াম উদ্বেগ এবং বিষণ্নতার লক্ষণগুলি মোকাবেলায় সহায়তা করতে পারে। ১০. ** নিজেকে শিক্ষিত করুনঃ** আপনার অবস্থা বুঝতে পারলে আপনি মানসিক স্বাস্থ্যের নিয়ন্ত্রণ নিতে সক্ষম হবেন। মনে রাখবেন, আপনার ব্যক্তিগত পরিস্থিতির উপর ভিত্তি করে পরামর্শের জন্য মানসিক স্বাস্থ্য পেশাদারদের সাথে কথা বলা অত্যন্ত গুরুত্বপূর্ণ।

Legend: **Highlighted text** indicates the detection of Anxiety-Depression Comorbidity of HealLM

In Table 4.6, the example shows that HealLM shows more structured than the MedGemma 1.5 4B model in that it gives more systematic, evidence-based, and comorbidity-aware answers. MedGemma 1.5 provides general recommendations for anxiety and depression, while HealLM does the same but also determines which instance of comorbidity is relevant to the case, shows how the knowledge-graph is constructed through clinical recommendations, adds more (local and external), and provides a systematic answer with a disclaimer. Specifically, the introduction of relationship-based reasoning is illustrated by the symptoms of insomnia which can show how HealLM works not just from the point of view of providing a response but also showing how the symptoms work for both anxiety and depression. Hence, in cases were finding the evidence, identifying the comorbidities, and applying relationship-based reasoning are important to address the mental health inquiry, HealLM architecture has much better transparency and context support than MedGemma alone.

4.3 Results & Discussion

In the following section, results from the testing procedures done by the proposed HealLM system will be presented and interpreted. The test was conducted using the developed test dataset and the metrics discussed in subsection 4.1 to evaluate the system in several aspects including response quality, semantic similarity, classification accuracy, reasoning regarding comorbidities, safety measures, and multilingual capabilities. First, the results will be presented quantitatively with further explanation of the results and the performance of the retrieval, ontology, and generation elements.

4.3.1 Results

The quantitative and functional evaluation result of the proposed HealLM system is shown below. The results are illustrative of its overall performance in the generation of results, semantic matching, classification and key mental health reasoning capabilities.

4.3.1.1 Response Generation Performance

Using ROUGE-1, ROUGE-2, ROUGE-L, and METEOR, the lexical resemblance between the generated replies and reference answers is calculated. The ROUGE-1 method looks at unigram matching, while ROUGE-2 pays attention to bigram matching. On the other hand, ROUGE-L identifies the longest matching subsequence. METEOR adds an extra level of similarity to the assessment by including word alignments and matching on the linguistic level. As a result, this system produced the following findings in Table 4.7:

Table 4.7: Response Generation Performance

Metric	Score
ROUGE-1	0.209
ROUGE-2	0.064
ROUGE-L	0.141
METEOR	0.288

ROUGE-1 score, which measures unigram overlap between the generated and reference answers, has received a score of 0.209. The ROUGE-2 score is lower than the ROUGE-1 score, reporting a value of 0.064 and indicating that the produced answers do not include phrases that originally appeared in the reference text.

The METEOR score of 0.288 proves that the generated text is lexically similar to the reference text. Overall, it appears that the artificial intelligence system can produce answers that are lexically similar yet able to produce answers with the help of machine learning.

4.3.1.2 Semantic Similarity Performance

Just evaluating the words used will not make it possible to find out if the answer provided regarding mental health is acceptable or not, because there can be two useful answers

worded very differently. That is why HealLM must be evaluated in terms of similarity between the resulting answer and the designated response.

The numbers acquired by the system equal 0.696, indicating that there is a significant degree of semantic alignment between the answers obtained and the answers expected.

Semantic evaluation is especially important in the context of HealLM, since it merges evidence from the local mental-health evidence base and Knowledge Graph into one final answer.

4.3.1.3 Semantic-Match Classification Performance

To make a more detailed analysis of the verbal relations between the produced and reference responses, semantic-match evaluation is used. The pairs that have matched correctly are seen as positive matches, while the mismatched responses are seen as showing the opposite. After this, the similarity is measured, which makes it possible to identify matched pairs.

The resulting classification performance is shown below in Table 4.8:

Table 4.8: Semantic Match Classification Performance

Metric	Score
Accuracy	0.952
Precision	0.964
Recall (Sensitivity)	0.939
Specificity	0.965
F1 Score	0.951

The system shows 95.2% accuracy, signifying that semantic similarity can be used for differentiating between actual and mismatch responses. Moreover, the precision standing at 96.4% exemplifies the fact that a substantial majority of the responses which matched semantically were correct matches. Furthermore, the recall score of 93.9% denotes that the greater portion of the real semantic matches was detected. In turn, the specificity of 96.5% suggests good separation from the mismatched responses and the F1-score of 95.1% indicates a good balance between precision and recall.

It must be emphasized that these figures denote performance in terms of semantic matching rather than clinical diagnosis. In other words, the numbers indicate semantic similarity of generated answers as compared to the actual answers and not their efficacy in diagnosing anxiety, depression, or any other mental illness.

4.3.1.4 Anxiety–Depression Comorbidity and Bridge-Symptom Evaluation

Given the primary aim of HealLM to facilitate anxiety-depression comorbidity reasoning, assessment also examines the capability of the system to deal with the inquiries on the connections between Generalized Anxiety Disorder (GAD) and Major Depressive Disorder (MDD).

The Clinical Knowledge Graph offers representation of disorders along with the respective symptoms, risk factors, medication, evaluation scales, clinical results, biological mechanisms, and other clinical aspects. Shared or bridge entities create the network among GAD and MDD making it possible for the system to perform multi-hop reasoning and provide information required by the task.

In this regard, the goal of the bridge-symptom evaluation is to ensure that the system can collect and use relevant shared information rather than treating anxiety and depression as two independent conditions. The aim is not to develop an independent clinical diagnosis but to test the possibility for the system to detect and clarify the connection between the relevant clinical aspects.

Table 4.9: Bridge Symptom Analysis

Question	Answer	Classification
A patient has experienced excessive and difficult-to-control worry about several areas of life for the past 8 months, along with restlessness, muscle tension, and sleep difficulties. Over the last 2 months, they have also developed persistent sadness, loss of interest in previously enjoyable activities, fatigue, feelings of worthlessness, and poor concentration. What mental health conditions could best explain this pattern of symptoms, and what features support your conclusion?	Excessive worrying over various aspects of daily living accompanied by restless behavior, muscle tensions, and disrupted sleeping patterns suggest potential signs indicative of generalized anxiety order. Persistent depressive emotions coupled with diminished pleasure derived from engaging in earlier enjoyed pursuits point towards depression too.	Anxiety-Depression Comorbidity Generalized Anxiety Disorder (GAD), Major Depressive Disorder (MDD).

Table 4.9 shows the Bridge Symptom Analysis for Anxiety-Depression Comorbidity cases.

4.3.1.5 Safety & Response-Quality Testing

In addition to the similarity of responses, checking the system's ability to deal with sensitive mental-health issues is also performed. It is important to note that the safety

component indicates the presence of dangerous content, including requests for information concerning suicide, self-injury, or intention to injure other people. These situations must be treated differently than ordinary requests about mental health issues.

The evaluation of the safety component is performed through qualitative tests by presenting typical low-risk and high-risk queries and analyzing the system's capability to distinguish between them. In case of the high-risk situation, the system is not expected to generate possibly dangerous recommendations but rather to give the proper safety-oriented answer, including information on emergency and crisis assistance, if needed.

The qualitative evaluation of responses enhances the quantitative evaluation of responses since a good mental health system must not only provide adequate answers according to meaning but also ensure their relevance and safety.

Table 4.10: Safety Awareness for Self-harm or Suicidal cases

Question	Safety
I am feeling worthless and I am so anxious about my life. I feel Suicide would be the best option to get rid of this	If you are having thoughts of suicide or self-harm, please reach out for support right away. In the US, you can call or text 988 (Suicide & Crisis Lifeline) any time, day or night. If you are outside the US, please look up a crisis line in your country -- a directory is available at https://www.iasp.info/resources/Crisis_Centres/. If you are in immediate danger, please contact your local emergency services now. You do not have to go through this alone. Please also consider reaching out to a mental health professional or a trusted person in your life as soon as you can.

Table 4.10 shows the safety awareness for any kind of self-harm, kill, suicide or crime related queries.

4.3.1.6 Multilingual Response Testing

Through its translation architecture, HealLM allows interactions in both English and Bengali. Bengali input is processed through the translation layer before proceeding into English-focused retrieval and reasoning, and the output can be translated back to Bengali for the audience.

The multilingual functionality is verified using typical queries in both Bengali and English and checking if the meaning of the input remains unchanged while the provided answer is still relevant. Care must be taken with the terminology of mental health since

mistranslated terms referring to symptoms, disorders, or risks can affect future retrieval and answers generation.

Table 4.11: Translation Response System of HealLM

Question	Answer	Translated Answer
A patient has experienced excessive and difficult-to-control worry about several areas of life for the past 8 months, along with restlessness, muscle tension, and sleep difficulties. Over the last 2 months, they have also developed persistent sadness, loss of interest in previously enjoyable activities, fatigue, feelings of worthlessness, and poor concentration. What mental health conditions could best explain this pattern of symptoms, and what features support your conclusion?	Excessive worrying over various aspects of daily living accompanied by restless behavior, muscle tensions, and disrupted sleeping patterns suggest potential signs indicative of generalized anxiety order. Persistent depressive emotions coupled with diminished pleasure derived from engaging in earlier enjoyed pursuits point towards depression too.	দৈনন্দিন জীবনের বিভিন্ন দিকের উপর অতিরিক্ত উদ্বেগ, বিশ্রামহীন আচরণ, পেশী চাপ এবং ব্যাহত ঘুমের নিদর্শনগুলি সাধারণভাবে শঙ্কিত হওয়ার সম্ভাব্য লক্ষণগুলির পরামর্শ দেয়। দীর্ঘস্থায়ী হতাশার অনুভূতি যা পূর্বে উপভোগ করা অনুশীলনে জড়িত থাকার ফলে প্রাপ্ত হ্রাসকৃত আনন্দ সহ যুক্ত হয় তাও বিষন্নতার দিকে নির্দেশ করে।

Table 4.11 shows Multilingual responses for HealLM in both Bengali and English queries

4.3.2 Discussion

The evaluation methodology offers a detailed evaluation of the proposed HealLM system based on its quantitative response capability and functional capability. Lexical and semantic metrics can be used to evaluate the quality of the generated responses in various ways and classification-based metrics offer a general evaluation of response correctness. The functional test further aims to evaluate the system outside the scope of general question answering in the following areas: anxiety–depression comorbidity, bridge-symptom reasoning, Clinical Knowledge Graph processing, safety handling, and Bengali English interaction.

The evaluation design is also in line with the core goals of HealLM. Specifically, the bridge symptoms and comorbidity cases enable the proposed GraphRAG component to be investigated for relationship-aware reasoning, not just information retrieval. Likewise, safety and multilingual testing are important practical needs of a mental-health support system. These functional evaluations, however, should be viewed in the context of the test cases defined and should not be considered as evidence of independent clinical diagnosis.

4.4 Summary

In general, the testing and evaluation approach was geared towards obtaining a

multidimensional and comprehensive evaluation of the HealLM system, covering its quality of response, semantic capabilities, functional capabilities, reasoning about comorbidities, safety features, and multilingual interaction. The framework does not just look at the quality of the responses generated, it considers the overall performance of the entire mental-health GraphRAG system, encompassing information retrieval, knowledge-graph-based reasoning, evidence integration, response generation, and safety verification.

The two different types of evaluation metrics, the quantitative and functional, can be used together to provide a more complete picture of HealLM's performance in various mental-health queries. The system evaluates its responses for dealing with anxiety and depression, as well as their comorbidity; and its responses to bridging symptoms, safety-sensitive queries and Bengali English interactions. Hence, the outcomes of the evaluation in this chapter can be a systematic basis for determining the effectiveness of the proposed architecture, strengths and weaknesses, and the extent to which it meets the specific research goals of the research.

Chapter 5

Engineering Standards and Design Challenges

This chapter deals with the engineering standards applied in the making of the studied HealLM system, as well as main design issues faced in the process of its implementation. The chapter gives emphasis to software, hardware, and communication standards applicable to the system, as well as to other alternatives which may have their pros and cons considered.

5.1 Compliance with the Standards

The HealLM has several components such as document processing unit, hybrid information retrieval system, clinical knowledge graph, large language model, multilingual translation component and safety verification mechanism. Because of this, the standards for software development, computing infrastructure, data processing and communication should be applied when discussing the interoperability, stability, maintainability and proper functioning of this system in practice.

5.1.1 Software Standards

Software Architecture of HealLM is designed to be modular and maintainable, which allows for the development and evaluation of the preprocessing, embedding generation, BM25 retrieval, reranking, knowledge-graph reasoning, response generation, translation, and safety verification components separately. The system is based primarily on open-source libraries written in Python, as well as standardized interfaces with various models, enabling seamless integration amongst the individual parts.

The ISO/IEC 25010 is a software-quality framework that can be regarded as relevant. It assesses functional appropriateness, performance efficiency, compatibility, usability, reliability, security, maintainability and portability. One way to assess the software is by the metrics of the model, such as accuracy, precision, recall, F1-score, ROUGE, METEOR, cosine similarity. These measures are very helpful for the measurement of system performance but offer less insight into more general software-quality attributes. Thus, there is a need for more broad grounds for evaluating the comprehensive quality of the entire HealLM system, which the ISO/IEC 25010 framework offers.
Structured graph representations are employed in the implementation of the knowledge graph to represent the relationships between disorders, symptoms, medications, risk

factors and comorbid conditions. A conventional relational database could be used as an alternative. Relational is a well-established, efficient system for tabular data, and is widely supported, but modelling and querying multi-hop relationships can be more complicated. For HealLM, a graph-based representation is thus more appropriate, in which relationship-aware retrieval and reasoning about comorbidities play an important role in the proposed architecture.

5.1.2 Hardware Standards

The computational resources that HealLM needs include those that can handle embedding generation, embedding retrieval, reranking, knowledge-graph generation, and inference and fine-tuning of LMs. As the development and experimentation process relies on the use of GPU based computing, especially NVIDIA CUDA compatible GPUs, the computational time of operations on the transformer model can be significantly shortened by using GPU acceleration.

The other option is a CPU only system, which has the benefits of less hardware resources, simple deployment, and may be cheaper for less demanding inference. But for large transformer models and fine-tuning, CPU execution is much less efficient. Hence, it was decided to use GPU based processing, as it is more efficient in computing the proposed MedGemma-based architecture.

Another point of hardware consideration is memory capacity since it is important to have a large amount of memory capacity for the large language models, embedding models, and reranking models during inference and training. Quantization techniques like 4-bit quantization can achieve lower memory usage without compromising too much on the comparatively limited resources of the GPU. The higher the precision of the computation, the more numbers will be better, but the amount of memory needed will be much higher. Thus, quantization and parameter-efficient fine-tuning is a viable approach to maintain a balance between computation and performance.

5.1.3 Communication Standards

The standards for communication are relevant to HealLM since the system can communicate with external knowledge sources, and other software components using services provided over the network. Access to third party sources like PubMed/PMC and other medical information sources can be achieved via web service interfaces or API, and communication amongst the different system components can be achieved with standard HTTP/HTTPS-based protocols.

HTTPS uses encryption and is preferable to plain HTTP, as it helps to preserve information shared between the client and an external service. The transport-layer security that HTTP 2.0 provides is less than what it provides in the case of HTTPS, but it offers simple security, with less protocol overhead. So, if your system deals with potentially sensitive mental-health related queries, you need to use HTTPS.

A structured data interchange format, JSON, is used as a practical data-interchange format for exchanging structured information between components. XML offers good

structural features and a lot of legacy support but is usually more verbose and not as lightweight as XML for modern machine-learning pipelines. The human-readable, lightweight, Python-supported, and convenient representation of queries, retrieved evidence, knowledge-graph information, and model configuration makes JSON a better choice for HealLM.

API-based communication is preferable for external medical information retrieval, to uncontrolled web scraping. An API-based approach will deliver structured responses and is generally more consistent and maintainable. While direct web scraping can offer access to more information, it can also be impacted by changes in the layout of the webpage and poses other reliability and compliance issues. Standardized API or structured web-based access is preferred, if available.

5.2 Impact on Society, Environment and Sustainability

The creation of HealLM has implications that extend beyond its capabilities as a technology, notably in such areas as mental health support, social well-being, environmental sustainability, and the responsible use of AI. In this section the benefits and limitations of the proposed system are discussed, along with the ethical questions it raises and the steps that could support its sustainable development and usage.

5.2.1 Impact on Life

HealLM aims to offer readily available information about mental health and initial support to people with symptoms of anxiety, depression, and their co-occurrence. The system's ability to display information in both languages helps overcome language barriers, making it more accessible for users who might struggle with English. The system may also provide a user with an improved understanding of the common symptoms, risk factors, information during treatment, and coping strategies.

HealLM is designed to be an assistive information system only, and not to replace qualified mental-health professionals. If someone misunderstands the information that was generated, it may have an impact on their decision to seek care from a professional. So, reducing potential harm by safety verification, evidence-based retrieval, appropriate disclaimers, and clear communication of the limitations of the system is important.

5.2.2 Impact on Society & Environment

At a community level, HealLM could potentially help enhance access to mental-health information, especially for Bengali speakers or those who might not have easy access to mental-health resources. Its GraphRAG architecture allows it to pull relevant evidence and take advantage of the relationships between disorders, symptoms, risk factors, and comorbid conditions, which can help inform more contextually relevant responses.

This system can also help to alleviate some of the obstacles to accessing initial mental health information, such as geographical access, the availability of professionals, and the complexity of medical information. However, it should not be an alternative to professional mental-health services.

If we look at the environmental aspect of the training and inference of LLM, it will use up computational resources, which ultimately equates to energy. To meet this challenge, HealLM fine-tunes the parameters of the base model and quantizes the model to decrease the computational and memory cost compared to training the full parameters. Additional gains from efficient retrieval and reuse of indexed resources may also help to mitigate unnecessary computation involved during repeated queries. These methods can enhance computation efficiency while retaining the usability and practicality of the system.

5.2.3 Ethical Aspects

This is significant for ethical reasons, as HealLM is in the mental health field. This system can be used to input sensitive information about emotional distress, depression, anxiety, self-harm or other sensitive situations. Thus, privacy, safety, fairness, transparency and responsible use of AI-generated information should be considered when developing systems. The proposed system implements safety verification, which indicates high-risk queries, like suicide, self-harm, and harm-to-others, and performs appropriate safety-related operations. The retrieval of evidence and knowledge-graph information also helps alleviate reliance on model-generated information with no evidence. The system should also make it clear that it can only provide information and should not be used for diagnosing or replacing a clinical diagnosis or medical opinion. Potential bias is another ethical issue. Information on mental health can vary depending on the population, culture, language and clinical context. HealLM can interact in Bengali and English, so any translation mistake or inappropriateness of culturally appropriate interpretation can impact the quality of the responses. Thus, it is essential to use multilingual tests and to carefully assess the responses generated to detect these limitations.

5.2.4 Sustainability Plan

HealLM's sustainability plan centers around ensuring that the technical, computational, and knowledge assets of the system can sustain the system over time. Update the knowledge base and clinical knowledge graph periodically, based on reliable medical and research resources, to keep the system alive with the development of mental-health knowledge. Individually, the components of the system, including retrieval models, knowledge sources, translation models, or safety mechanisms, can be redesigned or updated without redesigning the entire system.

Quantized models, parameter efficient fine-tuning, efficient retrieval & caching, and selective knowledge graph reasoning are all computational sustainability approaches. These methods help minimize the waste of computational resources and help make future deployment more feasible on limited hardware. For major changes, regular evaluation should also be conducted to ensure that the changes that improve in one component do not have a negative impact on response quality, safety or multilingual performance.

HealLM can be further extended in the long-term by ensuring a continuous clinical resource, better Bengali mental-health data, revised safety rules, and periodic model assessment. This provides the scaffolding for a system that not only will work on deployment but will be maintainable, adaptable, computationally efficient and responsible for the rest of its life.

5.3 Project Management and Financial Analysis

To develop HealLM, it is necessary to handle tasks such as management of computing power, software tools, datasets, testing operations, and development time. Because this system is mostly a research project and a prototype, the financial analysis only looks at how much computing power is needed for system development, testing, and maintenance. A comparison is made between two possible budget plans: a low-cost academic plan and a cloud computing plan. The former enjoys cost advantages since it prefers to use available hardware and open-source tools.

5.3.1 Project Management

The HealLM initiative is carried out through the execution of several phases of development, which encompass requirements investigation, preparation of literature and various datasets, formation of knowledge base and knowledge graph, model development and fine-tuning, integration, testing, quality assessment, and preparation of the final documentation. Modular architecture makes it possible to conduct tests of separate components prior to their integration into the system.
The project management is based on prioritizing separate project components in accordance with their computational and development complexity. Wherever possible, open-source models, libraries, and publicly accessible medical resources are used to minimize the development costs.

5.3.2 Budget Analysis

Table 5.1: Budget Analysis for HealLM

Resource	Estimated Cost (BDT)
Existing Computer	0
Kaggle GPU	0
Open-source models & libraries	0
Internet & Data access	3000
Storage	4000
Software	0
Miscellaneous expenses	2000
Estimated Total	**9000**

Table 5.1: shows the expenses and estimated cost of HealLM. The primary benefit of this method comes from its very low financial costs, which means that it can be used for academic studies. The main drawback, however, is the necessity of having available and effective hardware. Training and large-scale experimentation may take longer if GPU resources are limited.

5.3.3 Budget Selection and Rationale

In this academic implementation of HealLM, the choice is made for the minimum-cost budget as some existing resources may be already used by HealLM, thus alleviating it of the need for new investments. The proposed option offers the opportunity to significantly decrease the costs in the context of the project, while still being sufficient for all stages of the project such as research, model fine-tuning, system integration, and testing.
On the other hand, the cloud-based alternative is also applicable for future implementations where extra requirements in terms of GPU power and scalability are present.

5.4 Complex Engineering Problem

The HealLM system that has been proposed consists of different interlinked technical and area-specific challenges that need thorough investigation and engineering judgment. The difficulty of the problem is categorized according to the defined categories of complicated engineering problem-solving based on such criteria as the knowledge needed, depth of the analysis, familiarity with the issues, existing engineering regulations and guidelines, and interrelation of system components.

5.4.1 Complex Problem Solving

The features of complex problem-solving that are related to the proposed HealLM system are shown in Table 5.2. The mapping identifies the engineering problem categories that are related to the development and implementation of the suggested mental-health GraphRAG process.

Table 5.2: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
√		√	√	√		√

EP1 – Depth of Knowledge

HealLM necessitates an extensive understanding of a variety of educational fields, including artificial intelligence in addition to information retrieval, natural language processing, and many other themes including the knowledge of GraphRAG, mental health concepts, language models, and multilingual processing. This background knowledge

includes k3 Engineering Fundamentals, k4 Specialized Knowledge, k5 Engineering Design and lastly k8 Research Literature to combine different activities involved in retrieval as well as comorbidity reasoning, and efficiency assessment. Chapter 3 Methodology represents Depth of Knowledge.

EP3 – Depth of Analysis

The process of HealLM consists of multiple activities including query processing, hybrid retrieval, reranking, knowledge graph reasoning, validation of evidence, and generating results. Evaluation of the system is conducted by using both quantitative and qualitative methodologies. Rationale: The system makes use of retrieval, reasoning, validation, and evaluation processes which necessitate thorough analysis for obtaining reliable performance of the system. Both chapters 3 and 4 along with Fig 4.1 represent Depth of Analysis

EP4 – Familiarity of Issues

The technologies used include BM25, embeddings, LLMs, knowledge graphs, and RAG. However, in the context of mental health question answering, specific issues such as anxiety and depression comorbidity, bridging symptoms, and safety come into the picture along with Bengali and English. The rationale for the project is in combining known technologies with challenges associated with the field of interest that necessitate having meaningful engineering decisions and adaptations. Section 2.3, and subsections 3.2.2, 3.2.4, 3.2.5, 3.2.11, and 4.3.1.5 represent Familiarity of Issues.

EP5 – Extent of Applicable Codes

Since HealLM deals with the sensitive field of mental-health issue, it requires consideration of the aspects such as software quality, privacy, security, AI safety, responsible AI, and health-information handling. This is important as the self-same requirements guarantee the development of the system considering all required technical, safety and ethical aspects. Chapters 3 and 5 represent the Extent of Applicable codes.

EP7 – Interdependence

The main parts of HealLM, such as retrieval, reranking, reasoning via a graph of knowledge, validation of evidence, generation of MedGemma 1.5 4B [11], translation, as well as verification of safety are closely connected to each other. The main reason is that one part's performance in HealLM can determine the functioning of the whole system. Chapters 3 and 4 represent Interdependence.

Mapping with Knowledge Profile

Table 5.3: Mapping with knowledge Profile.

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
		√	√	√			√

In this part of the report, we will identify the relevant Knowledge Profile categories associated with the EP1 problem and the background to the problem. The development of HealLM will involve several knowledge areas, especially K3, K4, K5, K6, and K8, for the accomplishment of technical, design, implementation, and research processes related to the system in question.

K3 – Engineering Fundamentals

HealLM uses engineering principles in machine learning, natural language processing, information retrieval, data processing, and software engineering. The principles form the foundation of retrieval, reasoning, generation, and evaluation components.

K4 – Specialist Knowledge

To carry out the proposed work, specialists trained in large language models, GraphRAG, knowledge graphs, hybrid retrieval, clinical data, mental health concepts, and multilingual NLP will be needed. Understanding these key points is critical to the success of the work focused on the anxiety-depression comorbidity and building the new application in that domain.

K5 – Engineering Design

To create and integrate key components of HealLM architecture including hybrid retrieval, reranking, clinical reasoning using a knowledge graph, validation of evidence, generation of MedGemma 1.5 4B [11], translation, and verification of safety, knowledge related to engineering design is important.

K8 – Research Literature

The purpose of research literature is to identify the existing methodologies and choose suitable ones for the development of models and techniques that form the knowledge base of mental health and compare the proposed methods with the works of others. The relevant literature provides background information necessary for the development of

GraphRAG architecture, representation of knowledge in the field of mental health, techniques of LLM tuning, and evaluation methods.

5.4.2 Engineering Activities

This section establishes cues for the comprehensive HealLM project related to various Complex Engineering Activities (EA1–EA5). The system in discussion concerns a variety of computational resources, interactions of numerous system components and stakeholders, innovatively combining GraphRAG and LLM technologies, consideration of social and environmental factors, and both traditional and new engineering techniques.

Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
√		√	√	√

EA1 – Range of Resources

HealLM needs a wide variety of necessary resources including GPU-based hardware, datasets about mental health, medical documents, embedding and language models, knowledge-graph records, software libraries, and sources of external medical information. Reason: The coordination and integration of various resources confirm the large needs of the process of engineering. Sections 3.2.1–3.2.13 and Sections 4.1.1–4.1.5 represent EA1.

EA3 – Innovation

HealLM is the product of an innovative combination of existing technologies in specialized engineering solutions. It integrates hybrid textual retrieval, clinical Knowledge Graph reasoning, multi-hop reasoning, bridge-symptom representation, a customized MedGemma 1.5 4B model, Bengali English translation, evidence verification, and safety assessment focused on the topic of anxiety–depression comorbidity. In this unified architecture, the mental health question-answering task is studied instead of these technologies being deployed separately. Sections 2.3, 3.1.2, Sections 3.2.2–3.2.11, and Figure 3.1 represent EA3.

EA4 – Consequences for Society and Environment

In terms of social benefits, HealLM aims to promote access to mental health information, especially for Bengali-speaking individuals, as well as raising caution regarding the threats posed by inaccurate AI-provided data and the environmental aspect of extensive computation during model training and inference. Explanation:

As seen, these considerations entail various aspects such as safety, responsible utilization of AI, accessibility, confidentiality, and sustainability regarding computing. Sections 2.3, 3.1.2, Sections 3.2.2–3.2.11, and Figure 3.1 represent EA3.

EA5 – Familiarity

HealLM uses already-known technologies like natural language processing, BM25 technique, dense retrieve, transformer machines, knowledge graphs, and RAG, but using them in mental health comorbidity, safety-sensitive requests, and Bengali English interaction creates some unique challenges. Rationale: project mixes usual engineering technologies with very specific and new applications that need adaptation, testing, and evaluation. Sections 2.2–2.3, and Sections 3.2.2–3.2.11 represent EA5.

5.5 Summary

This chapter introduced the key engineering standards, social implications, project management issues and challenging engineering features that came with the creation of HealLM. Relevant software, hardware, communication standards and their alternatives and selection rationale were considered. The chapter also covered the project budget, alternative financial plan, and the potential revenue model, with a discussion of the impact of the system on life, society, environment, and ethical issues. Last, HealLM was mapped to the Complex Engineering Problem (EP1–EP7), Knowledge Profile (K1–K8), and Complex Engineering Activities (EA1–EA5), which are all interdisciplinary areas of knowledge that are required to build the proposed mental-health GraphRAG framework, analytic depth, innovation, resource demands, and system interdependence.

Chapter 6

Conclusion

The last chapter of this report is the concluding chapter regarding the design and assessment of the HealLM system. The chapter includes an overview of the contributions and results obtained by the project, describes the limitations of the system, and explains possible ways of enhancing and refining the system in the future.

6.1 Summary

The proposed mental-health GraphRAG system, called HealLM, was designed as a comprehensive mental-health system to aid in answering questions about anxiety, depression, and anxiety–depression comorbidities. The overall goal of the system is to integrate the information retrieval capabilities of RAG with the relationship-aware reasoning capabilities of a clinical knowledge graph and the natural language generation capabilities of a domain-adapted large language model. This pairing offers a framework to extract the appropriate mental-health data and offers answers supported by pertinent evidence.

In addition, the system includes a mental-health knowledge base that is filled with information from suitable medical and research materials. HealLM adopts a hybrid retrieval approach: dense semantic retrieval with BM25-based lexical retrieval, to improve the retrieval quality. The retrieved candidates go through the reranking process to get the highest-ranked proof. This retrieval architecture supports such a system to accept queries with semantically related meaning as well as queries containing specific clinical terminology.

One of the significant parts of the proposed framework is Clinical Knowledge Graph (CKG) that captures relationships between disorders, symptoms, risk factors, treatments, medications, and other clinical concepts. The knowledge graph is especially useful in tackling the research topic related to anxiety–depression comorbidity. The graph allows for concepts with different relationships to both conditions to be analyzed, in addition to direct text similarity, such as bridge or shared symptoms. This gives extra context which may help with question answering in relation to the comorbidities.

MedGemma 1.5 4B [11] is the generation component, which has been fine-tuned to the mental-health question-answering task at hand. Textual evidence and information from the graph is retrieved and added to the generation process to give context to the language model. Additionally, an evidence validation mechanism is integrated to evaluate the information available to support the final responses before they are generated. The system created is made to lessen dependence on the deep knowledge of models and make it easier to ground the answers published.

In addition, HealLM's safety verification layer tackles the delicate nature of mental-

health interactions. Specific questions related to potentially risky situations, such as suicide, self-harm and harm to others, are given extra safety consideration. The system is designed to give appropriate safety-oriented responses but still acts as an assistive information system and not as an autonomous clinical diagnostic system. This distinction is significant because information generated should not be used as a substitute for professional assessment or treatment of mental health problems.

The other important feature of the system is Multilingual Interaction between Bengali and English languages. The embedding of NLLB-200 [12] allows Bengali queries to be fed into the English based retrieval and reasoning system, and the generated response to be returned in Bengali when necessary. This adds a layer of accessibility and makes the framework relevant to users who primarily communicate in Bengali, related to mental health.

A test data set of 570 cases across various categories of mental-health queries was used to test the proposed system. Various aspects of system performance (classification oriented, lexical similarity, semantic similarity and functional capabilities) were evaluated. The quantitative evaluation resulted in an accuracy of 95.2%, a precision of 96.4%, a recall of 93.9%, a specificity of 96.5% and an F1 score of 95.1%. The relationship between generated and reference responses were also assessed by using ROUGE-1, ROUGE-2, ROUGE-L, METEOR and cosine similarity, for complementarity to the above metrics, as indicators of response quality.

In addition to numerical evaluation, functional tests were conducted to evaluate the system's performance in handling queries related to anxiety, depression, comorbidity, bridge symptoms, knowledge-graph-based reasoning, situations that are sensitive to safety, and Bengali English interaction. The evaluations measure the entire architecture and are not just designed to measure the underlying language model.

In the end, HealLM design and testing prove that it is possible to create a system for mental health Q&A that uses hybrid retrieval, clinical knowledge graph reasoning, evidence validation, and LLM generation. The results show that the proposed architecture can fulfill the main goals of the project as it integrates relationship-aware reasoning for anxiety–depression comorbidities and offers a framework for system enhancement, wider clinical validation, and system growth.

6.2 Limitation

The present HealLM framework is focused mainly on anxiety, depression and their co-occurrence, however with a focus on relationships between overlapping clinical concepts and symptoms. Therefore, the knowledge coverage of it is affected by the scope and quality of the curated mental-health knowledge base and clinical knowledge graph. If the goal is to expand to more mental-health conditions, more domain-specific resources and knowledge representation would be needed.

This evaluation was based on a testing database consisting of 570 cases that offers a framework with which to assess the performance of the system in response to various types of queries. But clinical and contextual, linguistic and cultural variations can be

larger in real-life interactions between mental health professionals and their patients. Furthermore, safety handling and Bengali English interaction could be studied mainly by functional testing and could be further validated by experts and quantitatively.
The quality of the overall response is also influenced by the accuracy of the retrieval, knowledge graph, translation and language generation components. The quality of retrieval may affect the evidence available for the generation task and translation may also affect the understanding of clinical expressions in Bengali. Likewise, language-model outputs can generate answers that may need the evidence to be validated and the safety to be checked. As such, HealLM is designed as a mental-health information system assistive tool, and the product of HealLM is meant to aid in information access, not to take the place of the professional clinical assessment or diagnosis.

6.3 Future Work

HealLM can be further developed by adding more mental-health disorders, symptoms, treatments, and comorbid conditions to the knowledge graph and knowledge base. In addition, larger and more varied mental health datasets in the Bengali language and more complete expert validated evaluation of the system can be helpful to improve the system. Further research can be done on better multilingual retrieval and generation, minimizing translation, and better handling of Bengali clinical terminology. More sophisticated graph reasoning, adaptive retrieval, and better evidence-ranking methods also can be used to improve the retrieval and knowledge-graph components.
Another key area is the evolution of the wider clinical and safety assessment, encompassing expert assessment, robustness testing, hallucination analysis and evaluation of realistic high-risk scenarios. Future improved models may also implement larger or more powerful DDLs but maintain the efficiency brought by quantization and parameter-efficient fine-tuning. These upgrades may enhance HealLM to become a more accurate, scalable, reliable, and practical tool for the future research and application of healthcare information.

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